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(54) **CONTROL UNDER UNCERTAINTY USING
CONSTRAINED ZONOTOPES**

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(57) **ABSTRACT**

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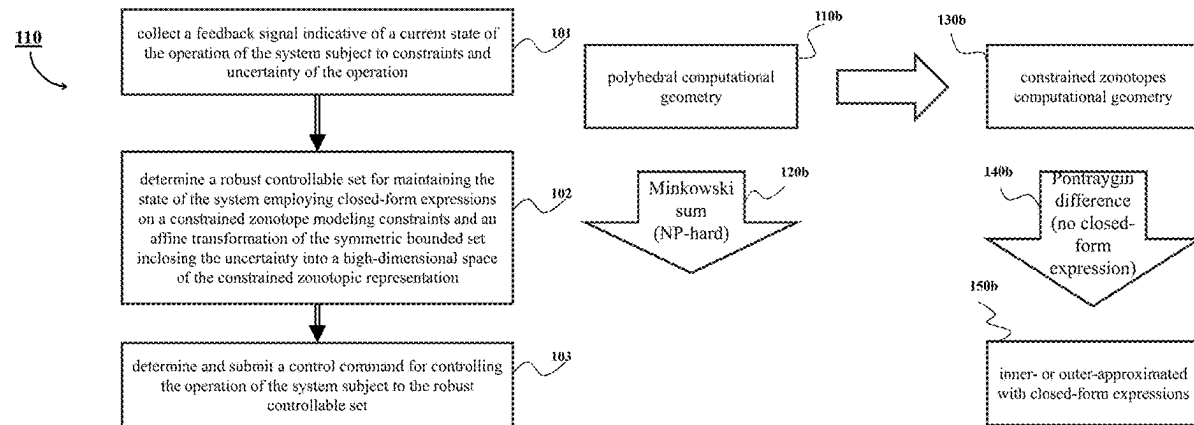
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A feedback controller collects a feedback signal indicative of the current state of the operation of the system subject to constraints and current uncertainty on the current state of the operation of the system and determines a robust controllable set for maintaining the state of the system employing closed-form expressions on a constrained zonotope defining the constraints on the state of the operation of the system and an affine transformation of the symmetric bounded set inclosing the current uncertainty into space of the constrained zonotope. The closed-form expressions include a closed-form approximation of a Pontryagin difference between the constrained zonotopic representation and the zonotopic transformation of the symmetric bounded set. The controller determines and submits a control command for controlling the operation of the system subject to the robust controllable set.



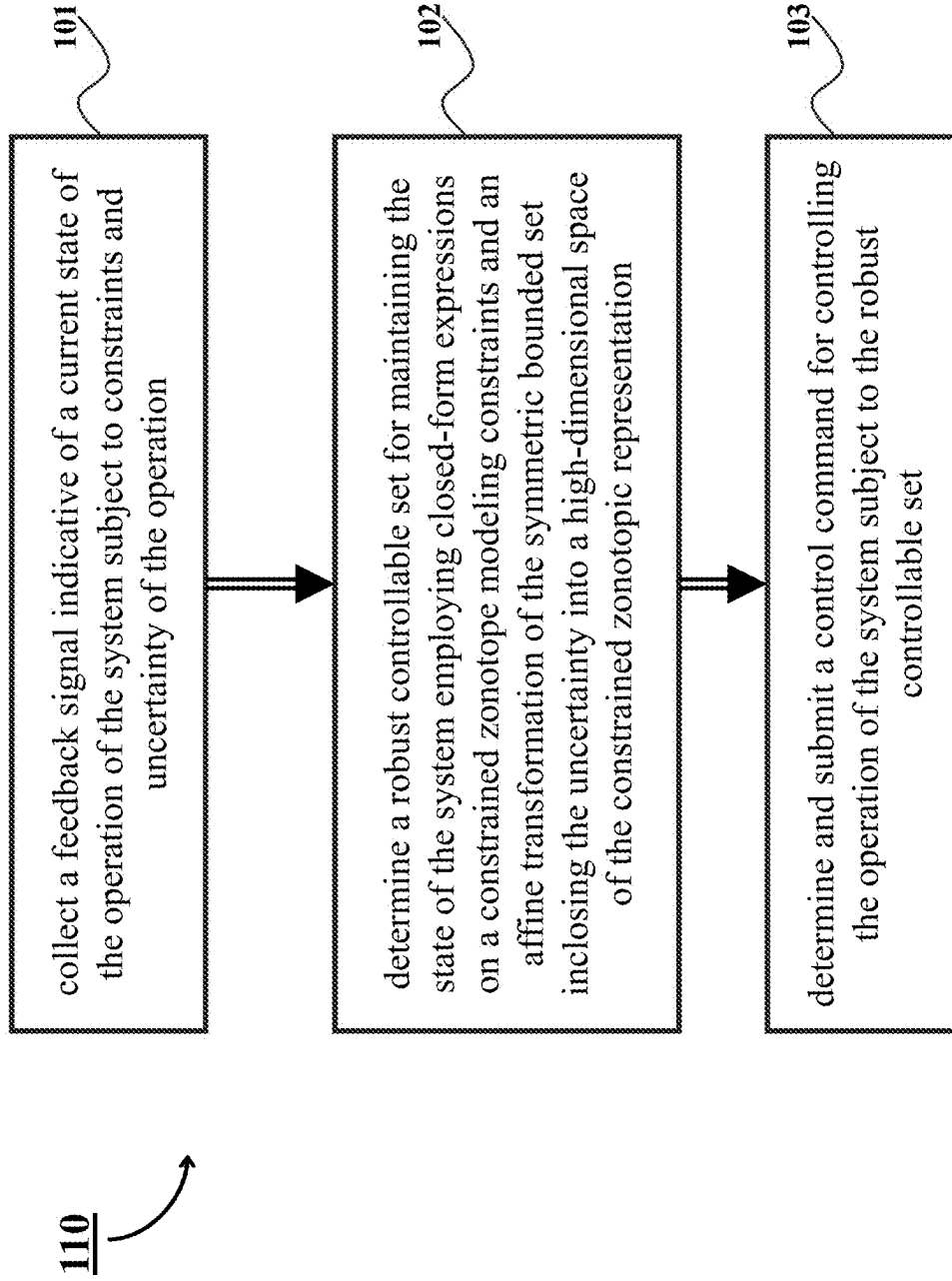


FIG. 1A

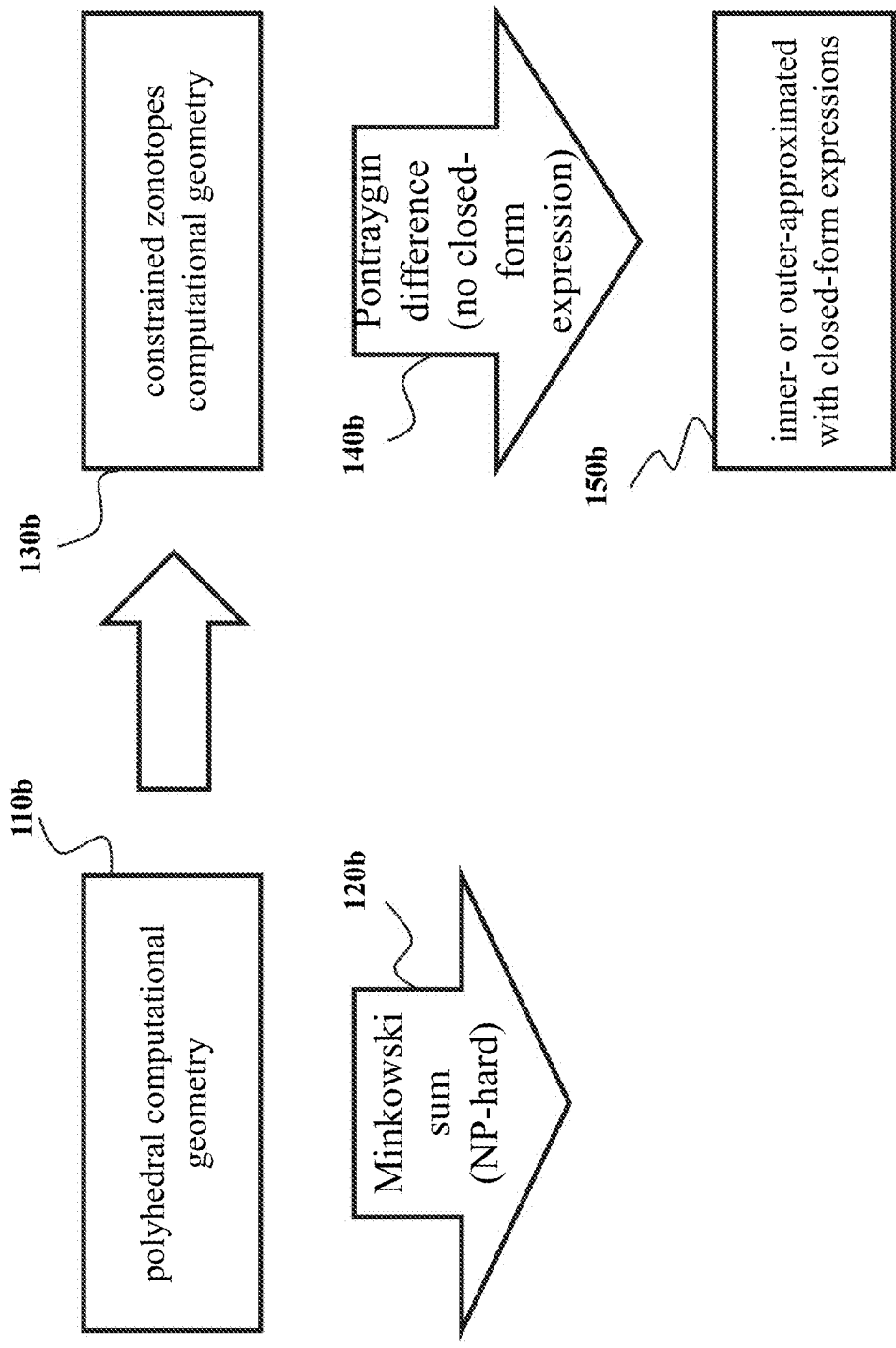


FIG. 1B

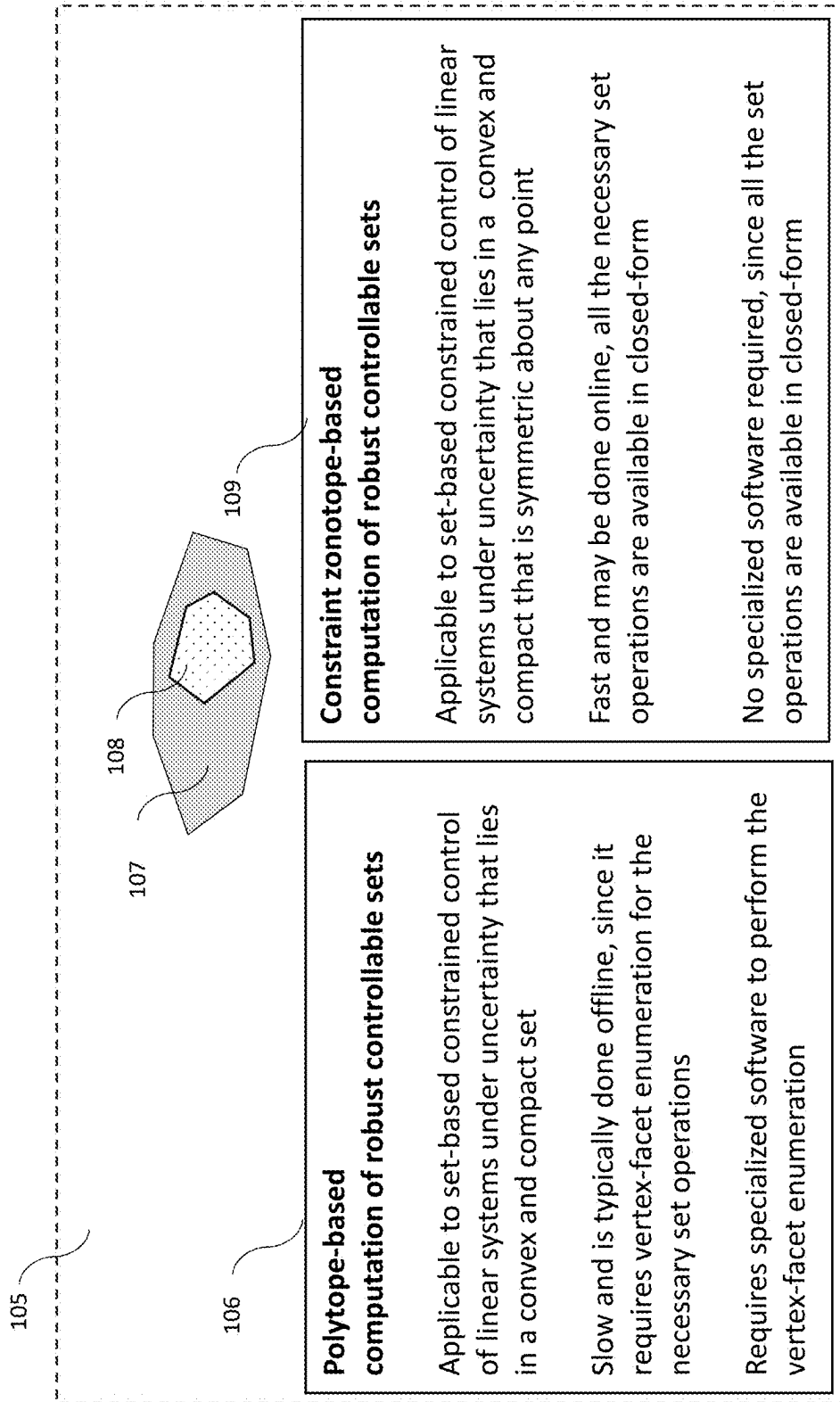


FIG. 1C

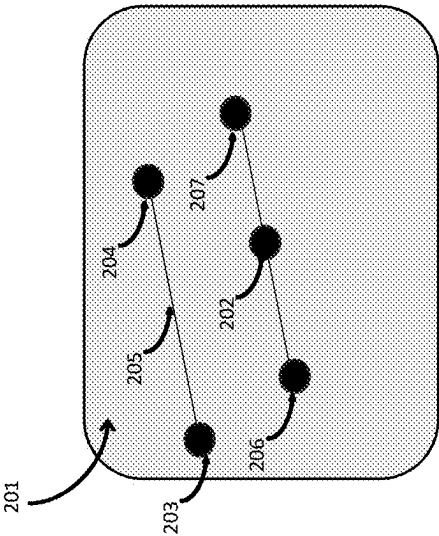


FIG. 2A

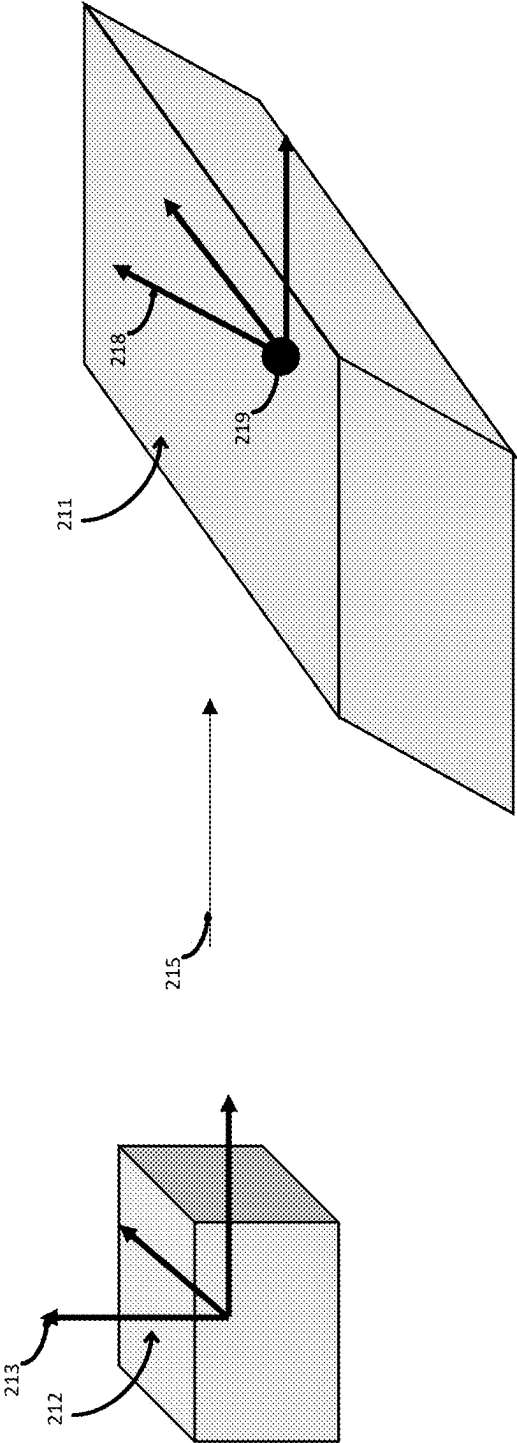


FIG. 2B

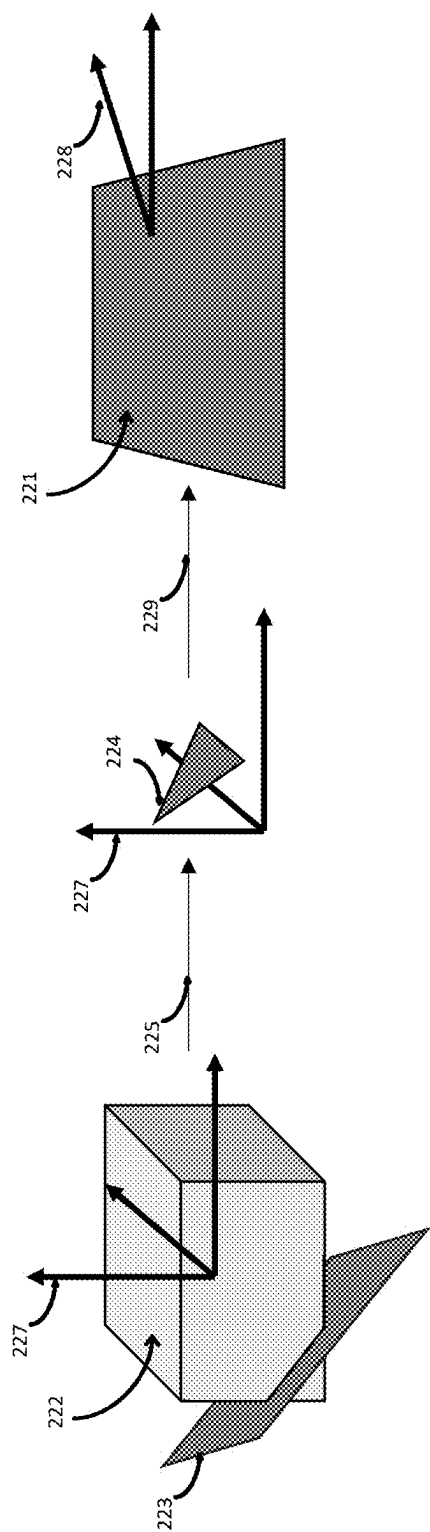


FIG. 2C

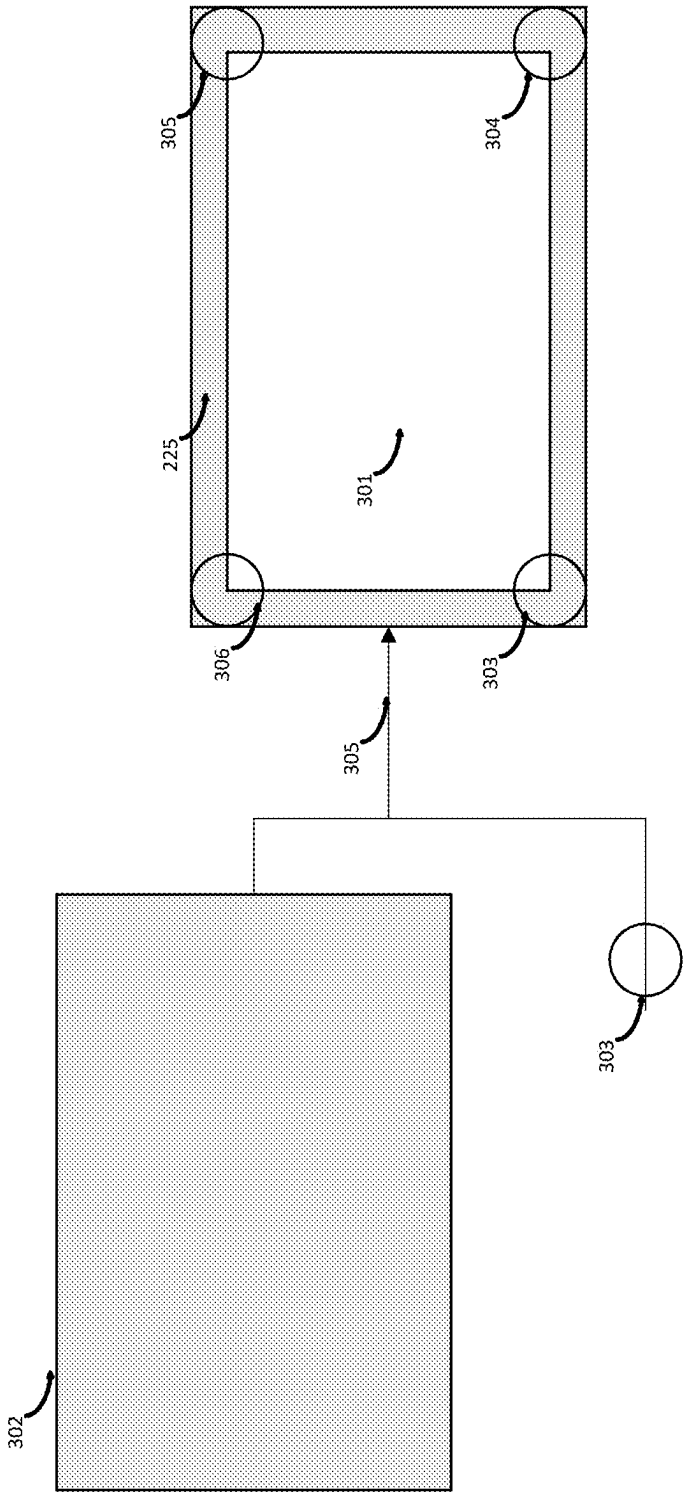


FIG. 3A

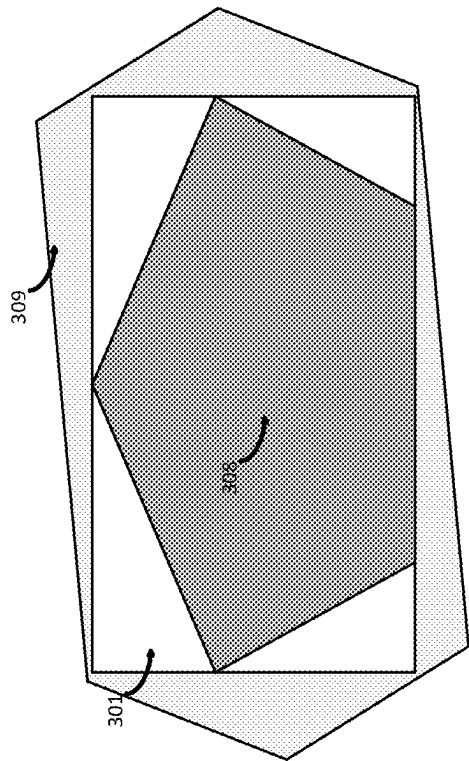


FIG. 3B

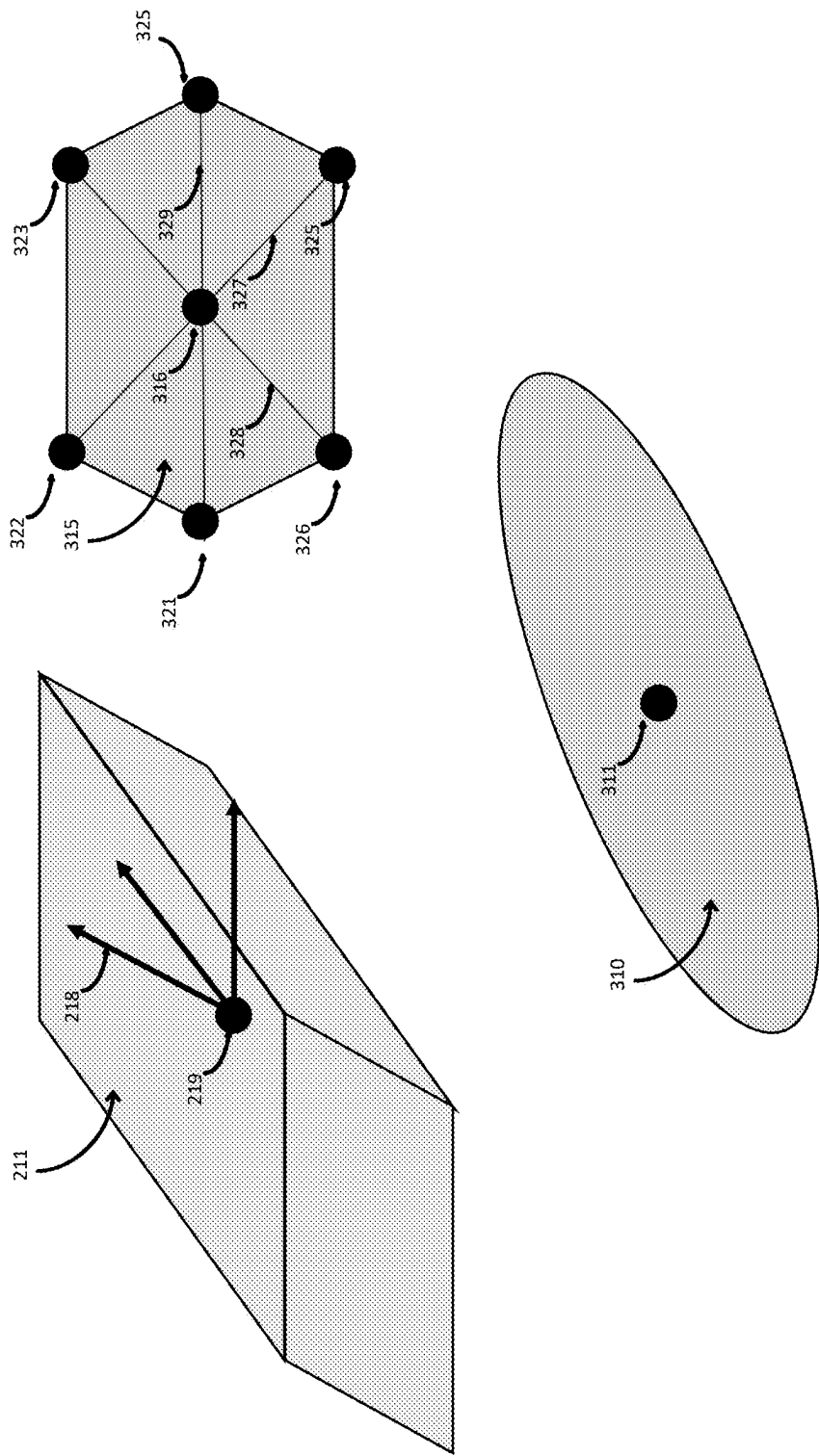


FIG. 3C

Algorithm 1 Inner-approximation of $\mathcal{C} \ominus S$

Input: Minkend $\mathcal{C} = (G_C, c_C, A_C, b_C)$ that is full-dimensional and subtrahend S that is convex, compact, and symmetric about any $c_S \in \mathbb{R}^n$

Output: Constrained zonotope $\mathcal{M}^- \subseteq \mathcal{C} \ominus S$

- 1: $\hat{S}_0 \leftarrow S - c_S$
- 2: $\Gamma \leftarrow [G_C; A_C]^{\dagger} [I_n; 0_{M_C \times n}]$
- 3: $D \leftarrow \text{diag}([1 - \rho_{S_0}(e_1^{\top} \Gamma); \dots; 1 - \rho_{S_0}(e_{N_C}^{\top} \Gamma)])$
- 4: **if** $D_{ii} \geq 0, \forall i \in \mathbb{N}_{[1:N_C]}$ **then**
- 5: $\mathcal{M}^- \leftarrow (G_C D, c_C - c_S, A_C D, b_C)$
- 6: **else**
- 7: $\mathcal{M}^- \leftarrow \emptyset$
- 8: **end if**

FIG. 3D

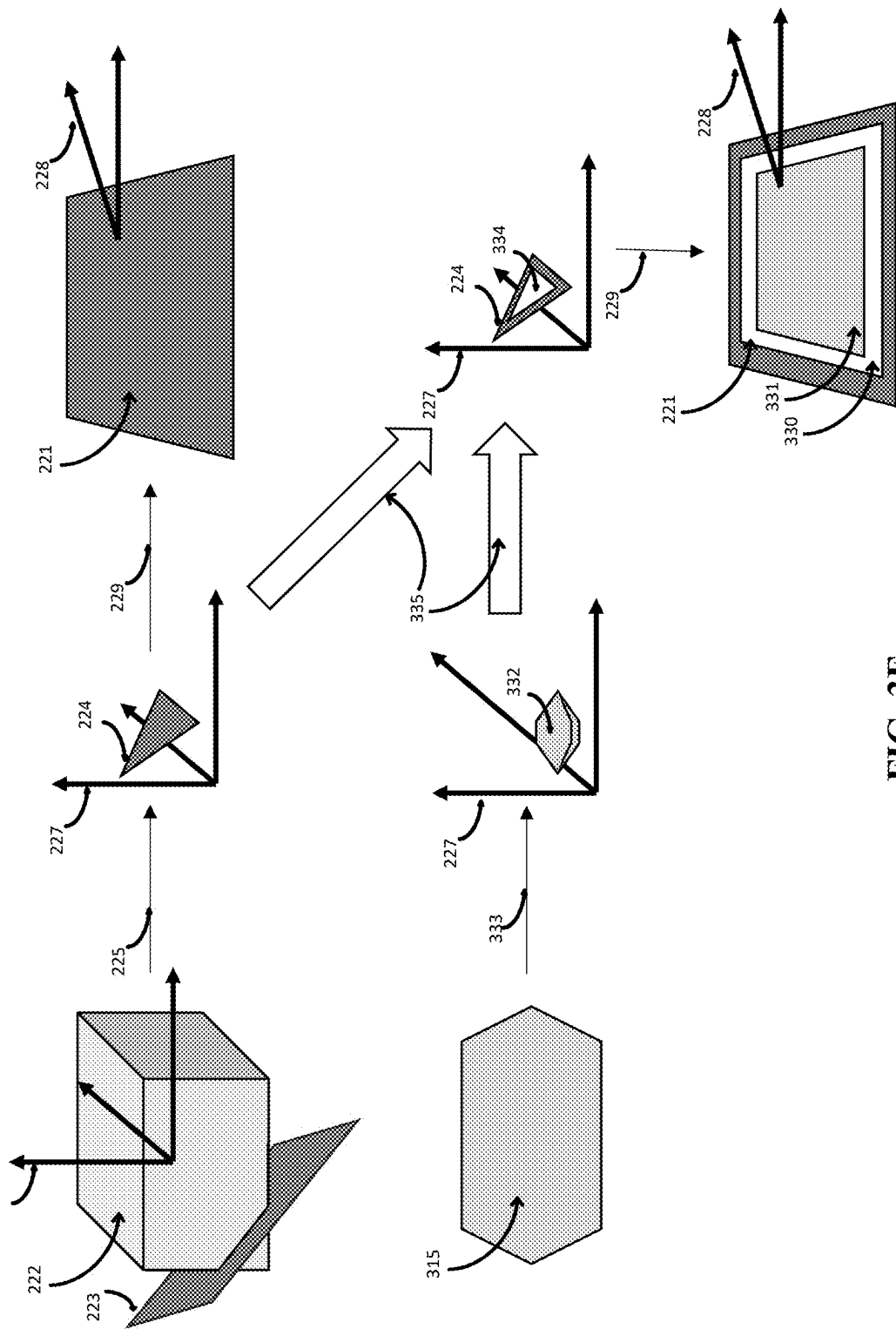


FIG. 3E

Algorithm 2 Convex polyhedral outer-approximation of a constrained zonotope $\mathcal{C}$

Input: Full-dimensional $\mathcal{C} = (G_C, c_C, A_C, b_C)$

Output: Convex polyhedron $\mathcal{P} = \{x : Hx \leq k\} \supseteq \mathcal{C}$

1: $V \leftarrow [v_1; v_2; \dots; v_{N_C}] \in \mathbb{R}^{N_C \times (n+M_C)}$ with

$$v_i = \frac{e_i^T [G_C; A_C]^T}{\|e_i^T [G_C; A_C]^T [G_C; A_C]\|_1}, \quad \forall i \in \mathbb{N}_{[1:N_C]} \quad (33)$$

2: $H \leftarrow [V; -V][I_n; 0_{M_C \times n}]$

3: $k \leftarrow \mathbf{1}_{(2N_C) \times 1} - [V; -V][-c_C; b_C]$

FIG. 4A

Algorithm 3 Outer-approximation of $\mathcal{C} \ominus \mathcal{S}$

Input: Minuend $\mathcal{C} = (G_C, c_C, A_C, b_C)$ that is full-dimensional and a convex and compact subtrahend $\mathcal{S}$ that is symmetric about any $c_S \in \mathbb{R}^n$

Output: Constrained zonotope $\mathcal{M}^+ \supseteq \mathcal{C} \ominus \mathcal{S}$

- 1: Compute a convex polyhedron $\mathcal{P} \supseteq \mathcal{C}$ using Algo. 2
- 2: Compute a convex polyhedron $\mathcal{M}_p^+ = \mathcal{P} \ominus \mathcal{S}$ where

$$\mathcal{M}_p^+ = \{x : H_P x \leq k_P - \lfloor \rho_S(h_1); \dots; \rho_S(h_{N_P}) \rfloor\}$$

- 3: Compute $\mathcal{M}^+ \leftarrow (\mathcal{C} - c_S) \cap \mathcal{M}_p^+$ using the fact

$$\text{that for any halfspace } \mathcal{H} = \{x : p^\top x \leq q\},$$

$$\mathcal{C} \cap \mathcal{H} = ([G_C, 0], c_C, [A_C, 0; p^\top G_C, d_m/2], [b_C; (q + p^\top c_C - \|p^\top G_C\|_1)/2]).$$

FIG. 4B

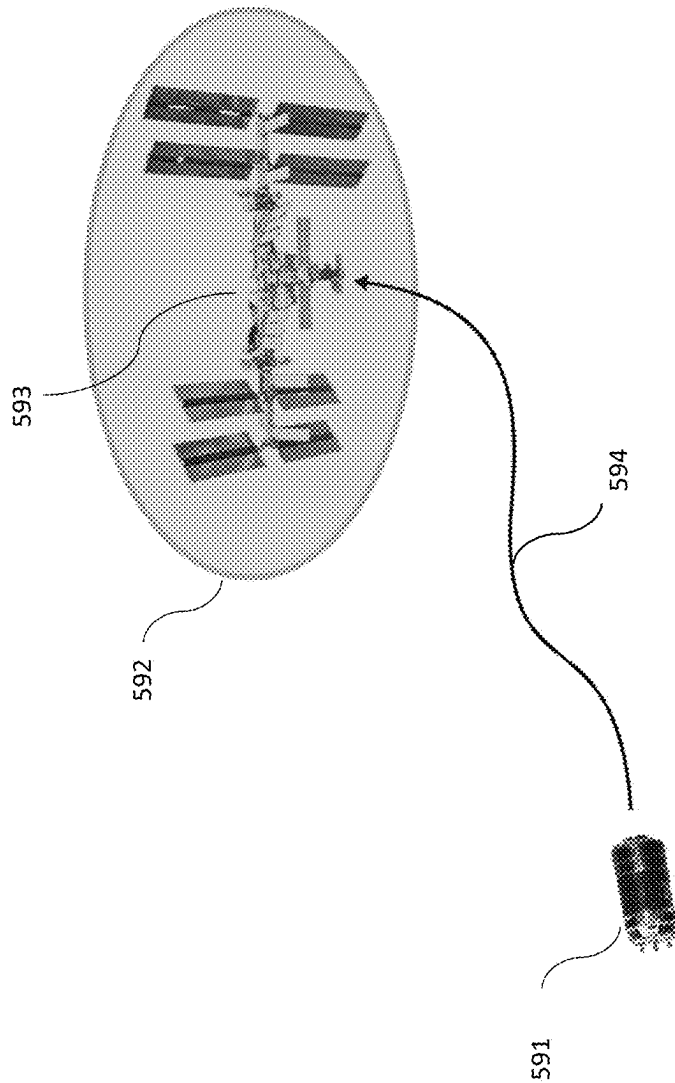


FIG. 5A

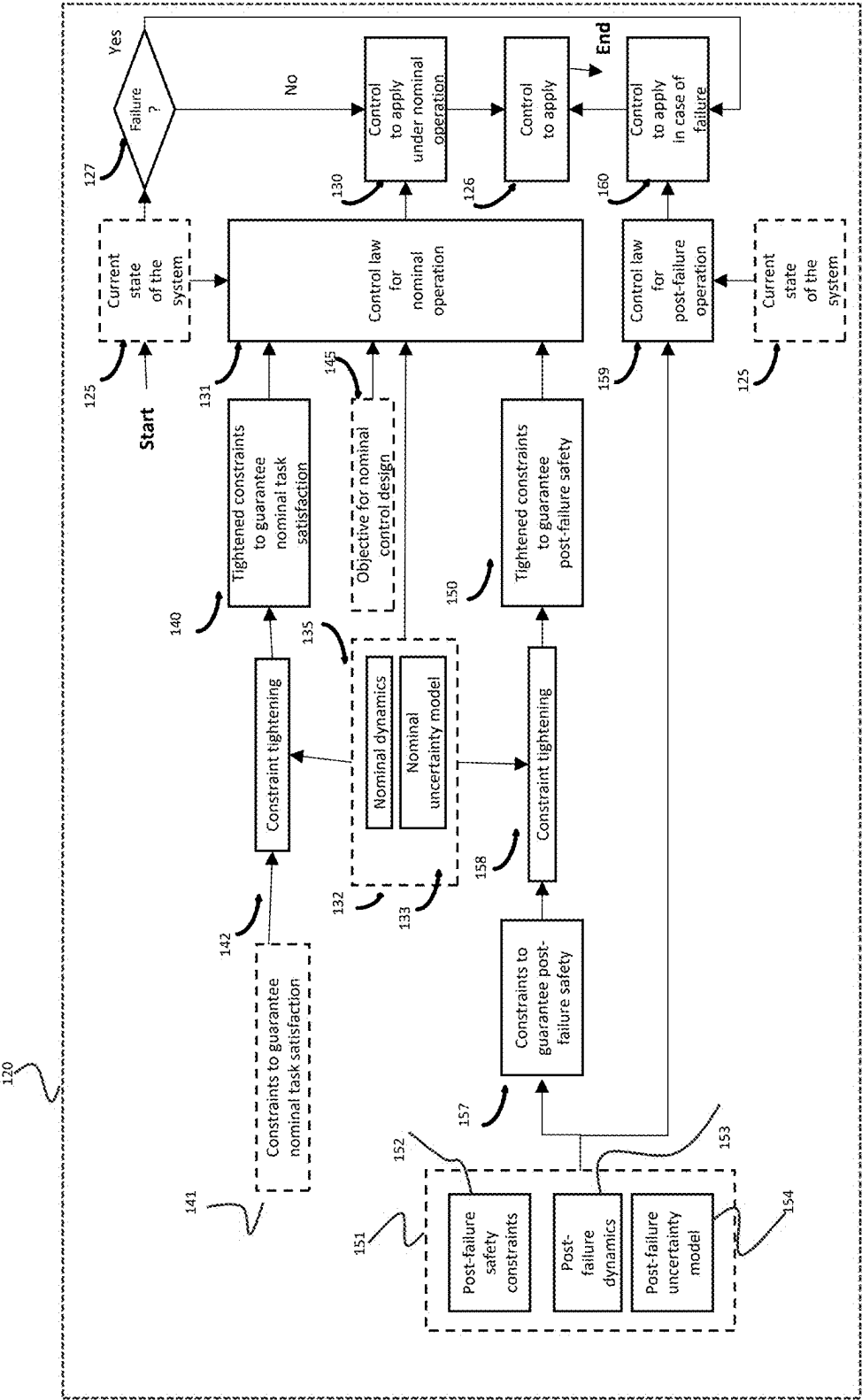


FIG. 5B

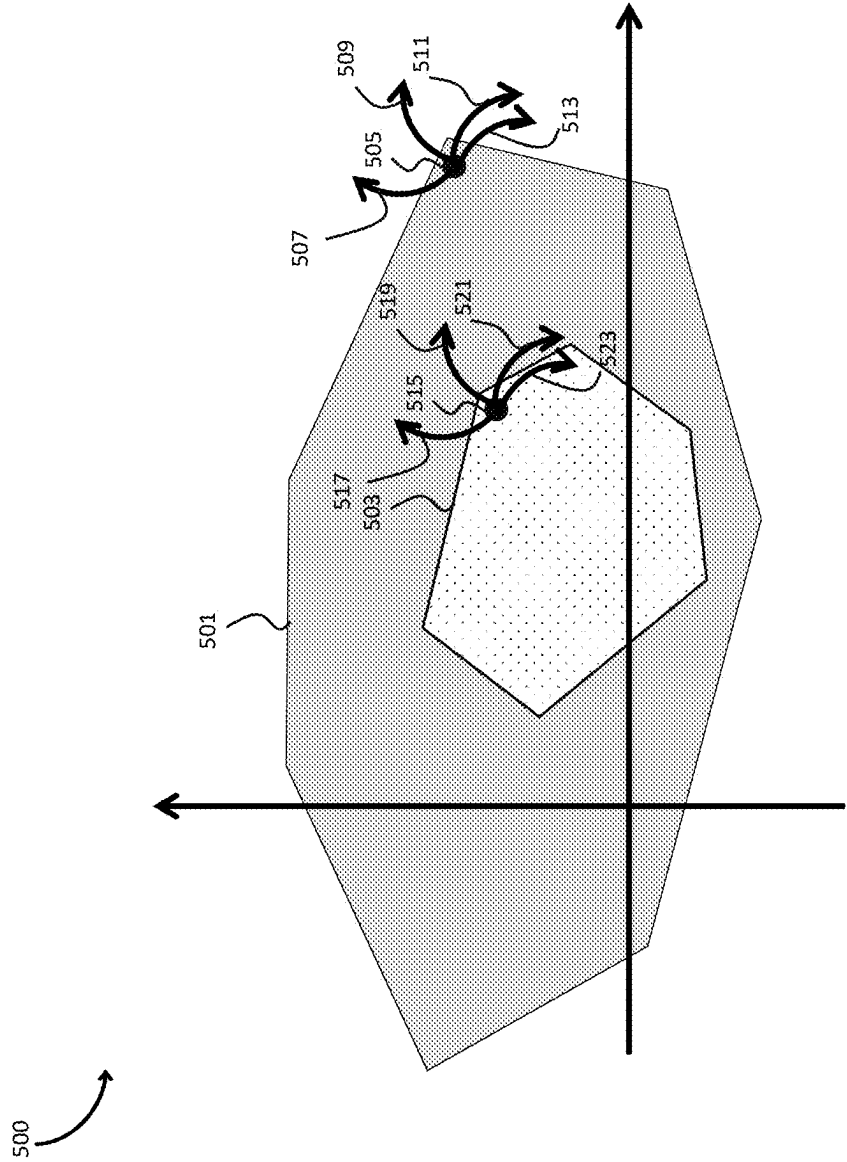


FIG. 5C

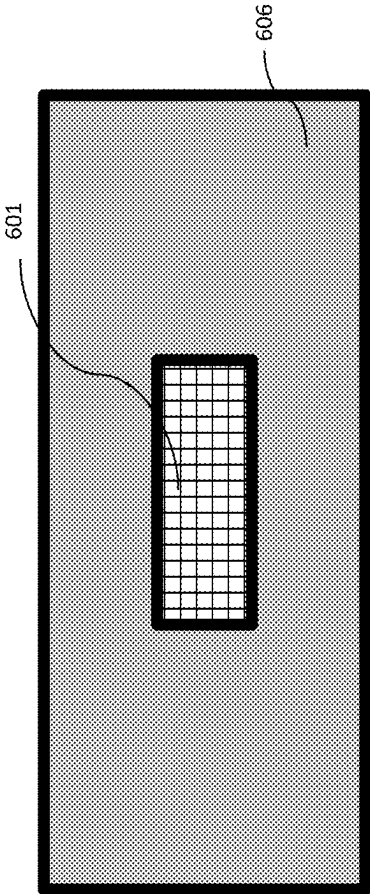


FIG.
6A

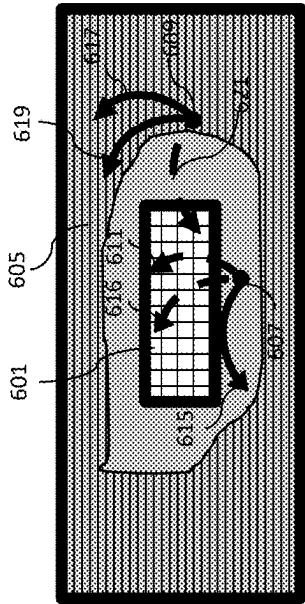


FIG.
6B

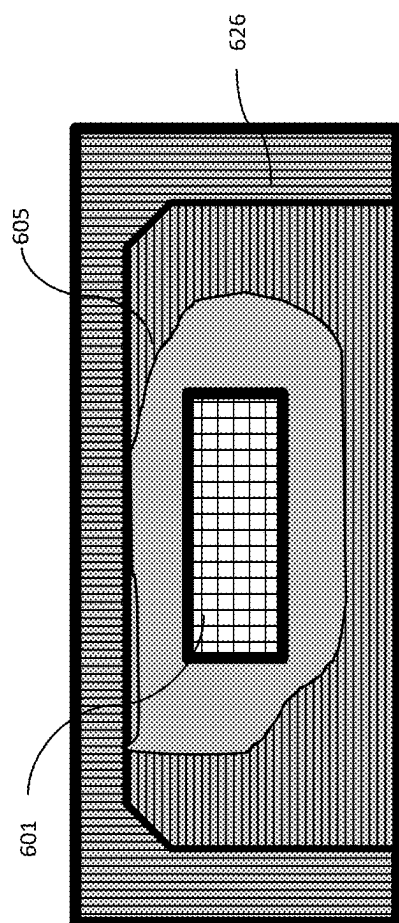


FIG. 6C

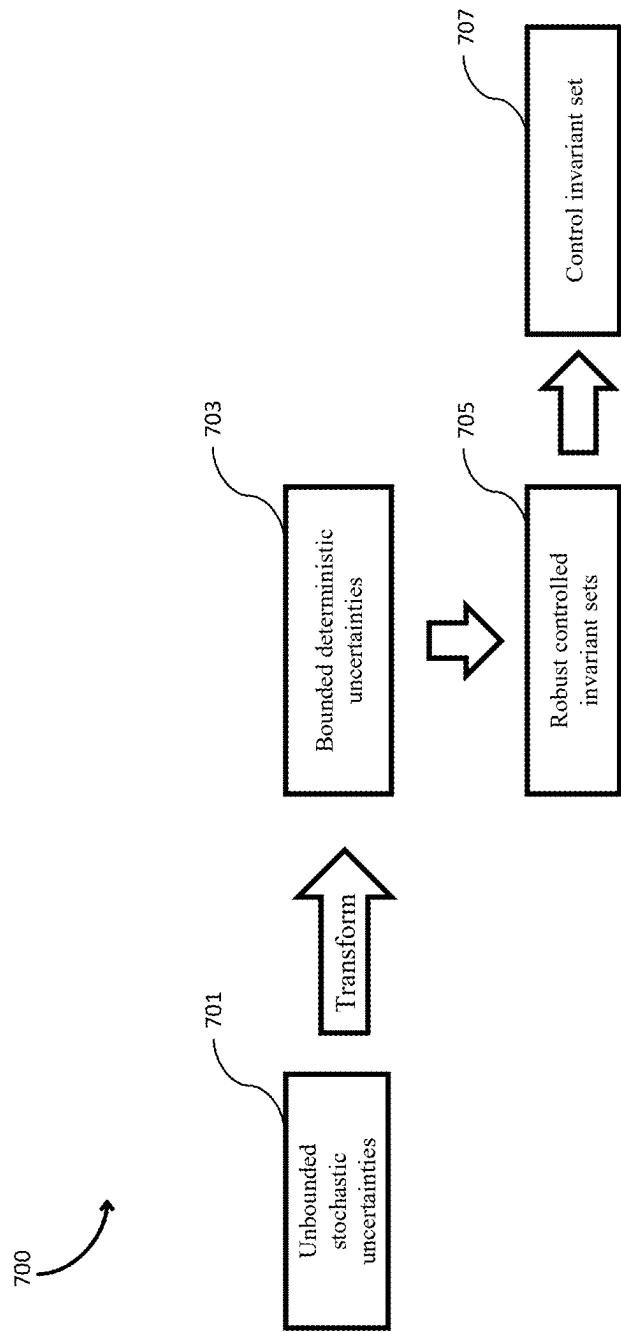


FIG. 7

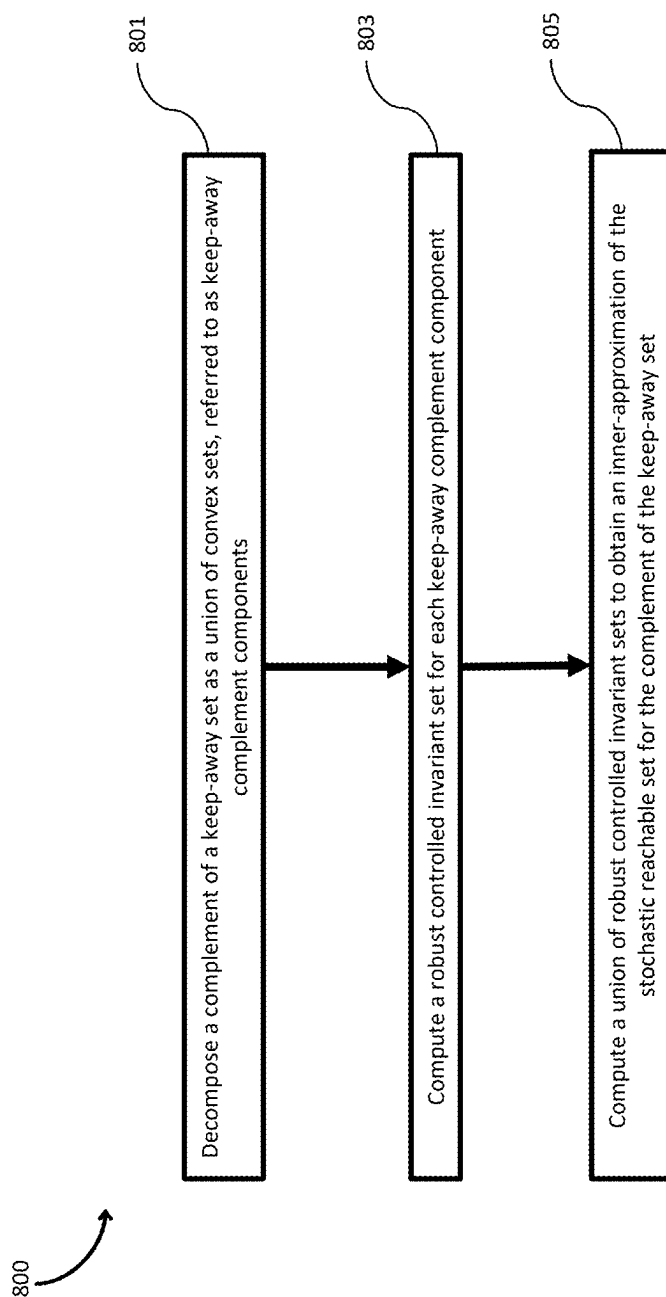


FIG. 8A

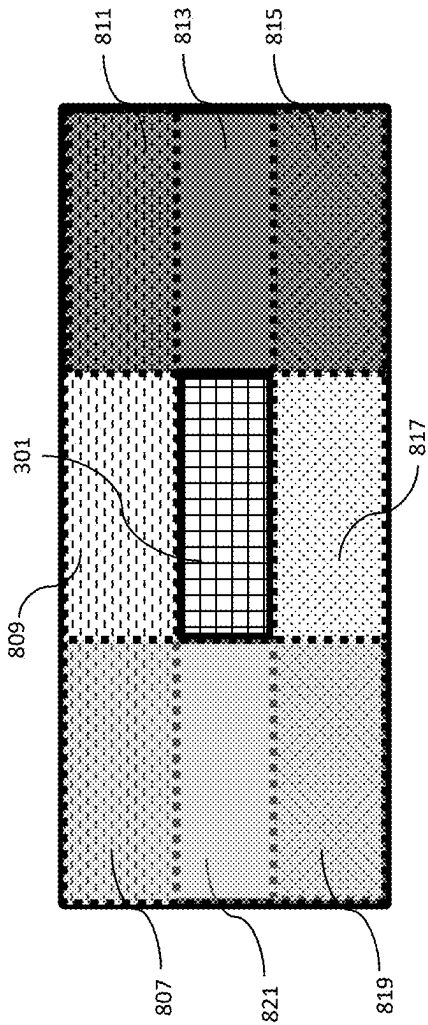


FIG. 8B

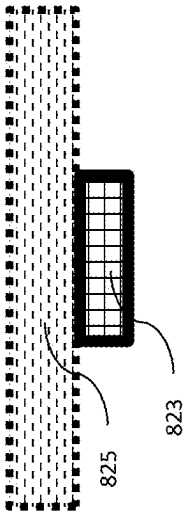


FIG. 8C

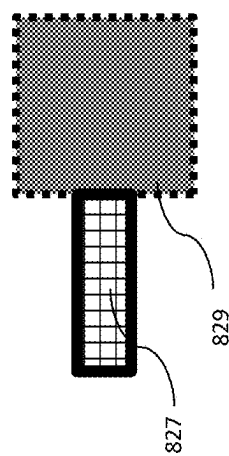


FIG. 8D

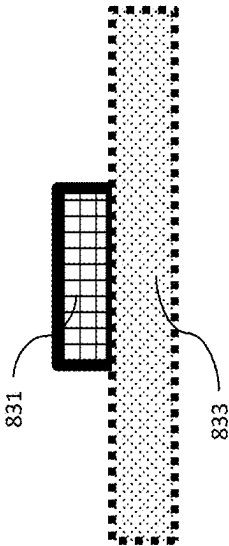


FIG. 8E

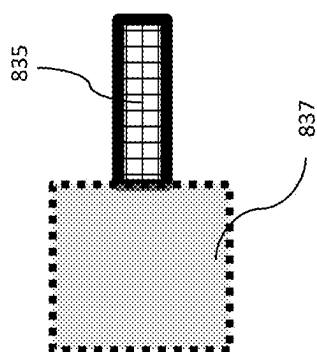


FIG. 8F

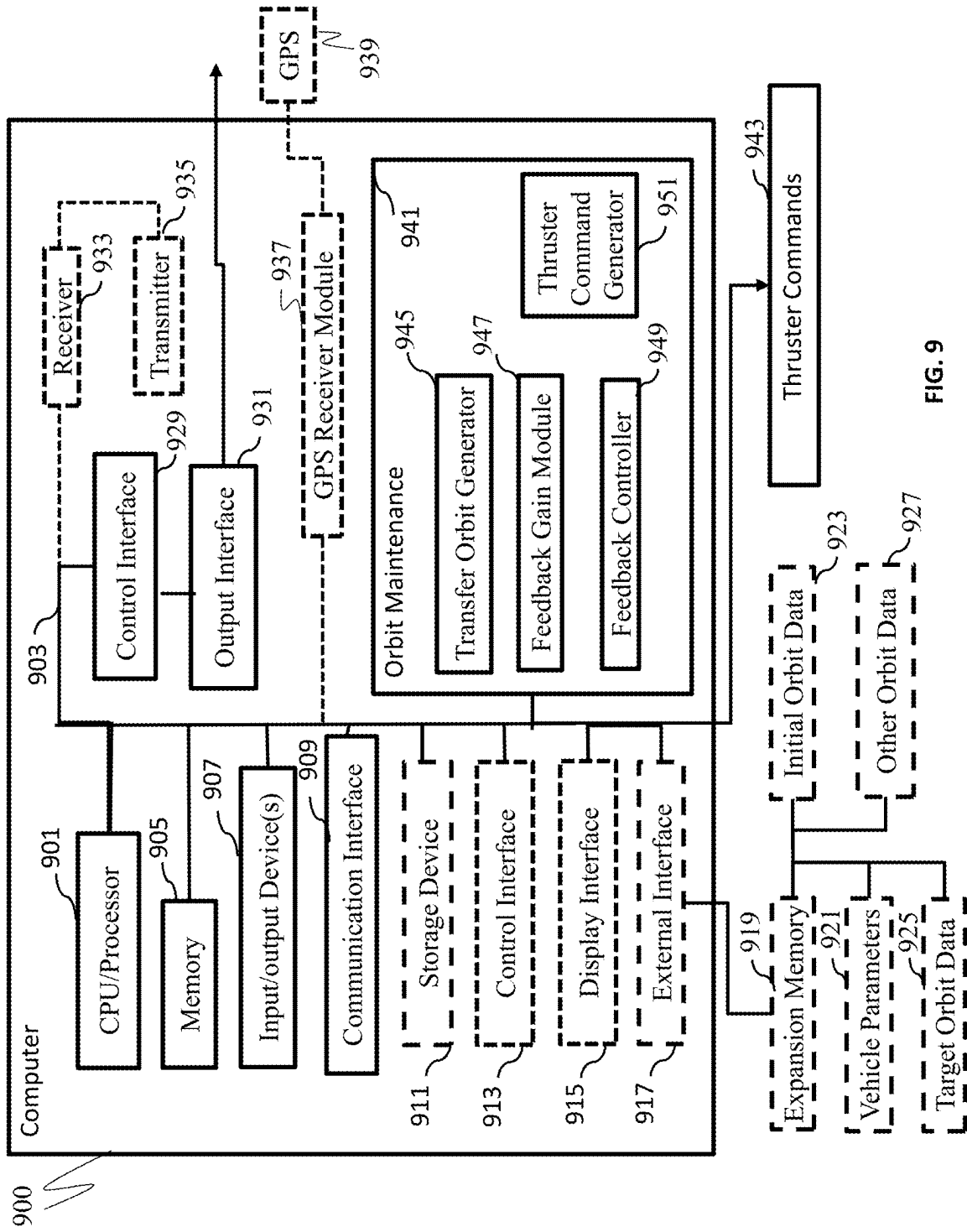


FIG. 9

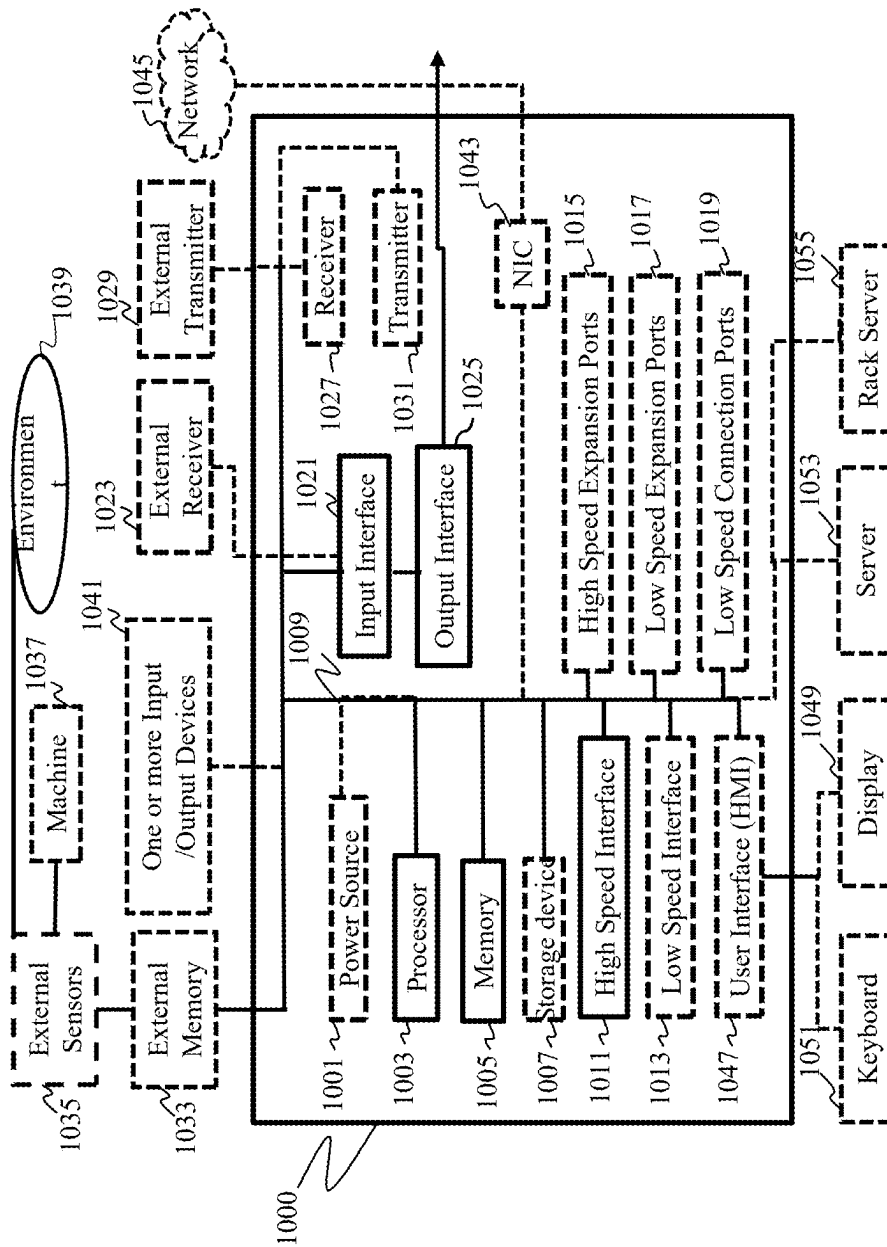


FIG. 10

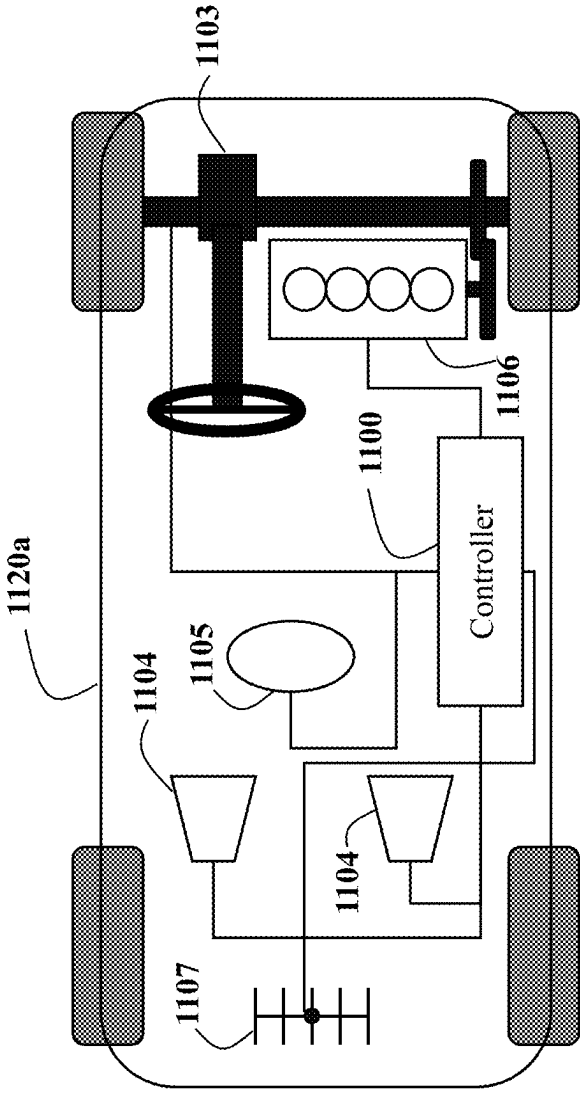


FIG. 11A

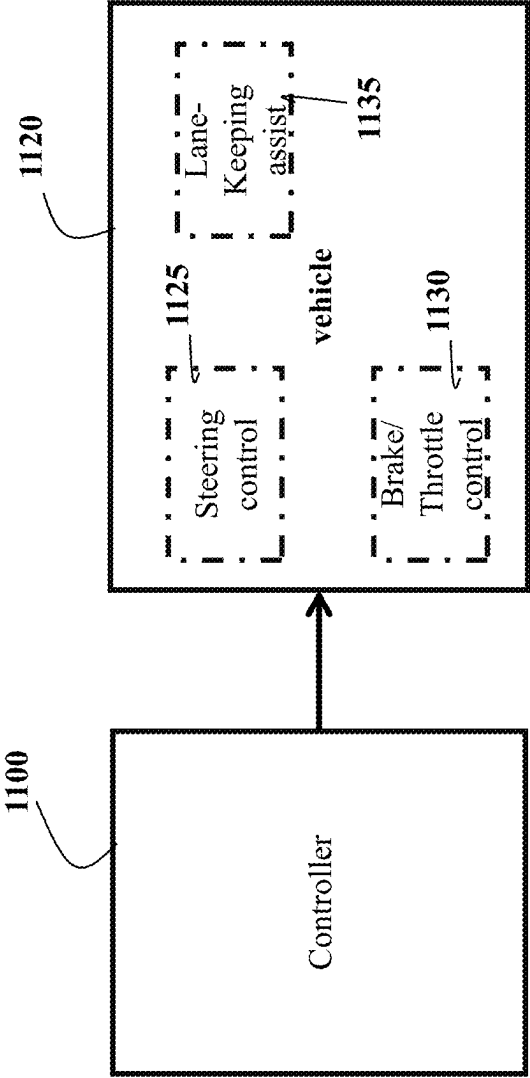


FIG. 11B

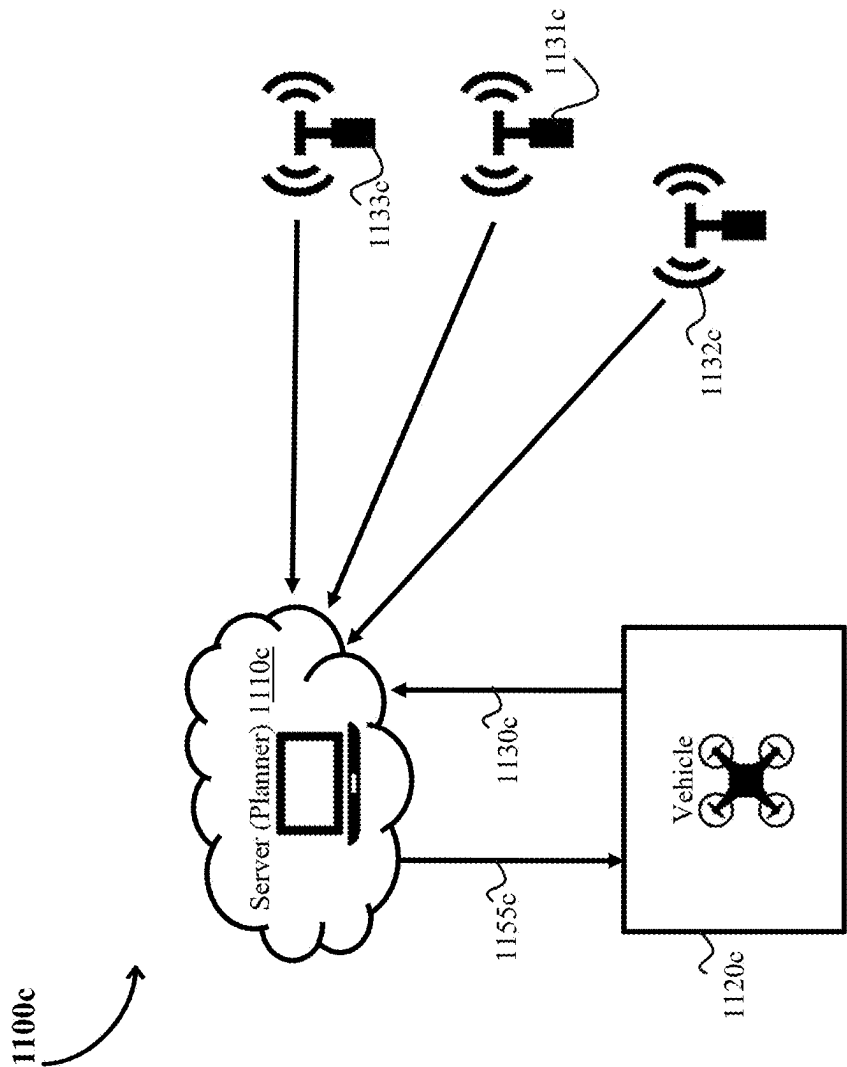


FIG. 11C

CONTROL UNDER UNCERTAINTY USING CONSTRAINED ZONOTOPES

TECHNICAL FIELD

[0001] The disclosure relates generally to control applications, and more particularly to a control system and a method for feedback control applications employing robust controllable sets.

BACKGROUND

[0002] Robust controllable (RC) sets, also known as robust backward reach sets, characterize a set of states from which a collection of possibly time-varying state constraints can be satisfied by a controlled state trajectory, under bounded control authority and bounded uncertainty. RC sets are used for robust model predictive control, fault-tolerant control, and verification of dynamical systems in a broad range of applications in space, transportation, and robotics. For discrete-time linear systems with additive uncertainty and bounded, polytopic state and input constraints, exact RC sets are polytopes, and the RC sets may be computed using set manipulations of polytopes. Polytopes are convex geometric shapes, and their manipulations have been well-studied in the literature. However, polytope-based RC set computation involves a projection that can be accomplished via vertex-facet enumeration, Fourier-Motzkin elimination, or multi-parametric optimization. These operations are either computationally prohibitive or cause numerical issues for high-dimensional system models arising from complex systems and/or in applications involving long safety horizons.

[0003] The RC sets can be computed using set operations defined in the field of computational geometry. Specifically, the robust controllable sets can be obtained via a set recursion involving set operations like intersection, affine transformation, Minkowski sum, and Pontryagin difference. Since several control applications involve linear dynamics and convex, polyhedral state constraints (characterized by a collection of linear constraints on the state), one can use polyhedral computational geometry to implement these recursions. In such an approach, it is possible to represent the state constraint sets as polyhedra, and then compute the robust controllable set by performing the above-listed set operations on polyhedra, for which there are well-known set computation algorithms. However, given one or more bounded polyhedra characterized by linear constraints, the Minkowski sum operation requires solving a vertex-facet enumeration problem, which is NP-hard. Consequently, existing approaches using polyhedral computational geometry suffer from numerical issues when used in high dimensions, and are limited to low-dimensional systems.

[0004] Therefore, there is still a need to compute RC sets in a computationally efficient manner suitable for online control applications.

SUMMARY

[0005] It is an object of some embodiments to provide a system and a control system and a method for feedback control applications including set-based constrained control and fault-tolerant control applications, employing robust and/or stochastic controllable sets using constrained zonotopes. Examples of such applications include control and space applications.

[0006] Robust controllable sets can be defined for a dynamical system with some bounded control authority subject to uncertainty lying in a bounded set. Here, the robust controllable sets characterize a set of states at the current time that, despite the uncertainty, can be steered to stay within a pre-determined sequence of state constraints over the next time steps. The robust controllable sets can be used in model predictive control to impose constraints on the states over the planning horizon so that some desirable phenomenon is guaranteed to occur even in the presence of disturbances. For example, using a suitably defined robust controllable set, some embodiments can ensure that the system degrades gracefully by staying in a pre-determined set of safe sets when needed or leaving open the possibility of satisfying some secondary requirements at any point in time.

[0007] Some embodiments are based on the recognition that robust controllable sets can be computed using polyhedral computational geometry. Specifically, the robust controllable sets can be obtained via a set recursion involving set operations like intersection, affine transformation, Minkowski sum, and Pontryagin difference. However, given one or more bounded polyhedra characterized by linear constraints, the Minkowski sum operation requires solving a vertex-facet enumeration problem, which is known to be NP-hard. Consequently, existing approaches using polyhedral computational geometry suffer from numerical issues when used in high dimensions, and are limited to low-dimensional systems.

[0008] Some embodiments are based on recognizing that constrained zonotopes provide an alternative representation of bounded polyhedrons (also called polytopes). A constrained zonotopic representation of a polytope characterizes the polytope as an affine transformation (projection) of the intersection of a unit hyper-cube in a high-dimensional space and an affine set. Affine sets are solution sets to some collection of linear equations. Constrained zonotopes provide closed-form, easy-to-evaluate expressions to compute the intersection, affine transformation, and Minkowski sum. However, no closed-form expression exists for the computation of the Pontryagin difference.

[0009] Some embodiments are based on recognizing that instead of considering the generic form of the Pontryagin difference, a computationally tractable approach for the Pontryagin difference between a constrained zonotope and a centrally symmetric, convex, and compact set is what is needed for many practical control applications. In addition, for these types of control applications, the Pontryagin difference can be inner- or outer-approximated with closed-form expressions.

[0010] Centrally symmetric sets satisfy the property that, for every point in the set, its reflection about a pre-determined point also belongs to the set. Convex sets satisfy the property that every line segment joining two points in the set is completely contained in the set. Compact sets are sets that are closed (including the boundary) and bounded. Examples of centrally symmetric, convex, and compact sets include zonotopes, ellipsoids, and convex union of intervals. An example of a control application with ellipsoidal uncertainty is the computation of RC set when the energy of the uncertainty is known to be bounded, in which case the uncertainty lies in an ellipsoid.

[0011] Various embodiments address the uncertainty of the control in a computationally efficient manner by inner-

and outer-approximating the Pontryagin difference between a constrained zonotope and a centrally symmetric set. For example, some embodiments compute the inner approximation to the Pontryagin difference by performing a specific scaling of the coordinate system in the high-dimensional space characterizing the constrained zonotope. The scaling that can guarantee an inner approximation has a closed form via a solution of a least-squares problem that utilizes the parameters characterizing the minuend (constrained zonotope) and the subtrahend (the centrally symmetric, convex, and compact set). Consequently, an inner approximation of the Pontryagin difference can be computed efficiently and accommodates sets defined in very large dimensional spaces as well.

[0012] In addition, some implementations determine the choice of scaling that coincides with the computationally expensive operation of lifting the subtrahend to the high-dimensional space associated with the constrained zonotope, performing the Pontryagin difference in the high-dimensional space, and then projecting it back to the low-dimensional space in which the subtrahend was originally defined.

[0013] Additionally or alternatively, some embodiments perform the outer approximation to the Pontryagin difference by computing a convex polyhedron that is guaranteed to contain the constrained zonotope. The description of the polyhedron is also available in closed form using convex optimization duality and least squares. The outer-approximation can be then determined by computing the Pontryagin difference between the polyhedral outer-approximation and the subtrahend.

[0014] Using the proposed inner and outer-approximation to the Pontryagin difference, some embodiments inner- and outer-approximate the robust controllable set for high-dimensional linear systems with polyhedral state constraints, polytopic input sets, and centrally symmetric, convex, and compact uncertainty sets. An inner approximation of the robust controllable set provides a sufficient condition for a state to lie in the robust controllable set. The outer approximation helps in quantifying the amount of conservativeness in the inner approximation.

[0015] In other words, some embodiments are based on recognizing that (1) there is a closed-form approximation, e.g., an inner or outer approximation, of a Pontryagin difference between the constrained zonotopic representation and a transformation of the symmetric bounded set into a high-dimensional space of the constrained zonotopic representations, (2) such a closed-form approximation is sufficient, i.e., practical and/or accurate enough, for feedback control applications employing robust controllable sets; and (3) such a closed-form approximation allows to employ closed-form computational geometry for determining robust controllable sets with computational efficiency suitable for online control of various mechanical systems.

[0016] In addition, some embodiments are based on recognizing that stochastic controllable (SC) sets are defined similarly to robust controllable (RC) sets. The SC sets characterize a set of states from which a collection of possibly time-varying state constraints can be satisfied with a user-specified likelihood by a controlled state trajectory, under bounded control authority and stochastic uncertainty. The differences between SC and RC sets are twofold. First, SC sets accommodate stochastic (and possibly unbounded) uncertainty, whereas RC sets require the uncertainty to be bounded but do not require any stochastic information.

Second, while RC sets guarantee absolute constraint satisfaction, SC sets provide constraint satisfaction up to a user-specified likelihood. Like RC sets, SC sets are advantageous for stochastic model predictive control, fault-tolerant control, and verification of dynamical systems, and have been used in a broad range of applications in space, transportation, and robotics.

[0017] The SC sets can be inner-approximated using appropriately defined RC sets. Consequently, the existing approaches to inner-approximate SC sets suffer from the same challenges identified above, and some embodiments of this disclosure remedy these shortcomings in the computation of SC sets by computing an inner-approximation of the SC sets using constrained zonotopes.

[0018] It is also an object of exemplary embodiments to provide a system and a method for controlling the motion of a spacecraft in a multi-object celestial system while avoiding an unauthorized entry into a keep-away region during a normal and abnormal (post-failure) operation of the spacecraft. As used herein, the normal operation includes moving toward a target in the keep-away zone, and the abnormal (post-failure) operation includes one or a combination of a failure to receive authorization to enter the keep-away zone and a failure of at least one component of the spacecraft. This problem comes from the requirements of spacecraft motion and rendezvous of multiple spacecraft forming a multi-object celestial system to ensure additional safety in space collaboration. To that end, it is an object of one embodiment to control the motion of the spacecraft toward the keep-away region such that upon detecting the abnormal (post-failure) operation there is a possibility to move the spacecraft while avoiding the keep-away zone.

[0019] Some embodiments are based on a recognition that there are states of the spacecraft in the proximity of the keep-away zone that may unavoidably lead the spacecraft into the keep-away zone even if the authorization to enter is rejected based on any reason including a total or partial failure of propulsion of the spacecraft. As used herein, the state of the spacecraft includes the location of the spacecraft and at least one or a combination of a velocity and an acceleration of the spacecraft. Additionally, internal and external forces acting on the spacecraft during its motion, such as inertia and force of gravity, may change the state of the spacecraft to specific states that may lead the spacecraft into the keep-away zone, regardless of any efforts that the spacecraft may perform.

[0020] Some embodiments are based on the realization that this fault-tolerant control problem can be addressed by using a robust controllable (RC) set of values of the state of the spacecraft that the spacecraft should be within. As used herein, an RC set is determined such that when the state of the spacecraft is within a robust controllable set there is a control command produced by a control law that maintains the state of the spacecraft within a (possibly time-varying) collection of state constraints despite the controlled and uncontrolled, internal and external forces acting on the spacecraft. In such a manner, if the spacecraft is within the RC set, a control law within specified control limits exists such that the spacecraft does not enter the keep-away zone.

[0021] Some embodiments are based on recognizing that it is computationally expensive to compute the RC sets online, i.e., during real-time controlling of the spacecraft. Hence, the embodiments address this problem using a closed-form approximation, e.g., an inner or outer approxi-

mation, of a Pontryagin difference between the constrained zonotopic representation and a transformation of the symmetric bounded set into a high-dimensional space of the constrained zonotopic representations

[0022] Some embodiments are based on recognizing that requiring the motion of the spacecraft to lie in a constrained zonotope can be enforced by a collection of linear constraints. Consequently, some embodiments utilize off-the-shelf (convex) optimization tools to enforce the condition that the spacecraft belongs to a robust controllable set, that is inner-approximated by a constrained zonotope.

[0023] Additionally or alternatively, some embodiments are based on a recognition that requiring the motion of the spacecraft to lie in a plurality of constrained zonotopes can be enforced by a collection of mixed-integer linear constraints. Consequently, some embodiments utilize off-the-shelf (mixed-integer) optimization tools to enforce the condition that the spacecraft must belong to a robust controllable set, that is inner-approximated by a constrained zonotope.

[0024] Some embodiments are based on the understanding that stochastic uncertainties may be transformed into bounded deterministic uncertainties with a predefined likelihood specifying a range of values on the stochastic uncertainties. Enforcing that the spacecraft with stochastic state lies in an SC set with high likelihood can be guaranteed if the mean state of the spacecraft lies inside the Pontryagin difference of the SC set or its inner approximation and an ellipsoid characterized by the stochasticity of the set. Consequently, some embodiments utilize convex optimization tools and the disclosed systems and methods to compute the Pontryagin difference of a constrained zonotope to enforce the condition that the spacecraft must belong to a stochastic controllable set that is inner-approximated by a constrained zonotope with a user-specified likelihood.

[0025] Accordingly, an embodiment discloses a feedback controller for controlling an operation of a mechanical system subject to uncertainty on a state of the operation of the system, the feedback controller comprising: at least one processor; and a non-transitory memory having instructions stored thereon that, when executed by the at least one processor, cause the feedback controller to: collect a feedback signal indicative of a current state of the operation of the system subject to constraints and current uncertainty on the current state of the operation of the system, wherein the current uncertainty lies in a symmetric bounded set and includes one or a combination of uncertainty on dynamics of the system, uncertainty on a state of the system, uncertainty on control commands to the system, and uncertainty on constraints on the state of the operation of the system; determine a robust controllable set for maintaining the state of the system employing closed-form expressions on a constrained zonotope defining the constraints on the state of the operation of the system and an affine transformation of the symmetric bounded set inclosing the current uncertainty into space of the constrained zonotope, wherein the closed-form expressions include a closed-form approximation of a Pontryagin difference between the constrained zonotopic representation and the zonotopic transformation of the symmetric bounded set; determine a control command for controlling the operation of the system subject to the robust controllable set; and submit the control command to an actuator of the system causing a change in the state of the operation of the system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The presently disclosed embodiments will be further explained with reference to the attached drawings. The drawings shown are not necessarily to scale, with emphasis instead generally being placed upon illustrating the principles of the presently disclosed embodiments.

[0027] FIG. 1A shows a block diagram of a feedback controller **110** for controlling the operation of a mechanical system subject to uncertainty on the state of the operation of the system according to some embodiments.

[0028] FIG. 1B shows a schematic of some principles employed by different embodiments.

[0029] FIG. 1C shows an illustration of the benefits of the proposed method in computing robust controllable sets using constrained zonotopes computational geometry, according to an embodiment of the present disclosure.

[0030] FIG. 2A illustrates a convex and compact set that is centrally symmetric about point having properties employed by some embodiments.

[0031] FIG. 2B illustrates a zonotope employed by some embodiments.

[0032] FIG. 2C illustrates a constrained zonotope employed by some embodiments.

[0033] FIG. 3A illustrates the Pontryagin difference between two sets employed by some embodiments.

[0034] FIG. 3B illustrates an inner-approximation and outer-approximation of the set employed by some embodiments.

[0035] FIG. 3C shows various examples of sets that are convex, compact, and symmetric about some point having principles employed by some embodiments.

[0036] FIG. 3D shows pseudocode of an algorithm of a method for computing an inner-approximation of the Pontryagin difference according to some embodiments.

[0037] FIG. 3E illustrates the intuition behind the algorithm described in FIG. 3D used to compute an inner-approximation of the Pontryagin difference between a constrained zonotope and a convex, compact set that is symmetric about a point.

[0038] FIG. 4A illustrates an algorithm to compute the polytopic outer-approximation of a constrained zonotope according to some embodiments.

[0039] FIG. 4B shows pseudocode of an algorithm of a method for computing a constrained zonotopic outer-approximation of the Pontryagin difference between a constrained zonotope and a convex, compact set that is symmetric about a point according to some embodiments.

[0040] FIG. 5A illustrates the motion of spacecraft in a multi-object celestial system, according to an embodiment of the present disclosure.

[0041] FIG. 5B shows a block diagram of a trajectory design method for fault-tolerant applications that use set-based control according to some embodiments.

[0042] FIG. 5C shows an example of a two-dimensional projection of a robust controllable set corresponding to a constraint set, according to an embodiment of the present disclosure.

[0043] FIG. 6A shows a keep-away set and its complement, according to an embodiment of the present disclosure.

[0044] FIG. 6B illustrates a stochastic controllable set, according to an embodiment of the present disclosure.

[0045] FIG. 6C illustrates an inner-approximation of the stochastic controllable set, according to an embodiment of the present disclosure.

[0046] FIG. 7 shows a schematic for determining the robust controllable set by leveraging the robust controllable set, according to an embodiment of the present disclosure.

[0047] FIG. 8A shows a block diagram of a method for computing the inner-approximation of the stochastic controllable set, based on the robust controllable sets, according to an embodiment of the present disclosure.

[0048] FIG. 8B shows the complement of the keep-away set decomposed as the union of convex sets, according to an embodiment of the present disclosure.

[0049] FIGS. 8C-8F collectively show convex halfspaces that together define the complement of the keep-away set used by some embodiments.

[0050] FIG. 9 is a schematic diagram illustrating some components used for implementing the methods and the systems of the present disclosure.

[0051] FIG. 10 is a schematic illustrating by non-limiting example a computing apparatus for implementing the methods and the systems of the present disclosure.

[0052] FIG. 11A shows a schematic of vehicle including the controller, according to some embodiments of the present disclosure.

[0053] FIG. 11B shows a schematic of interaction between the controller and controller of vehicle, according to some embodiments of the present disclosure.

[0054] FIG. 11C shows a schematic of an embodiment where an unmanned aerial vehicle (UAV) is controlled using the controller implemented on a server, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0055] In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent, however, to one skilled in the art that the present disclosure may be practiced without these specific details. In other instances, apparatuses and methods are shown in block diagram form only in order to avoid obscuring the present disclosure.

[0056] As used in this specification and claims, the terms “for example,” “for instance,” and “such as,” and the verbs “comprising,” “having,” “including,” and their other verb forms, when used in conjunction with a listing of one or more components or other items, are each to be construed as open-ended, meaning that that the listing is not to be considered as excluding other, additional components or items. The term “based on” means at least partially based on. Further, it is to be understood that the phraseology and terminology employed herein are for the purpose of the description and should not be regarded as limiting. Any heading utilized within this description is for convenience only and has no legal or limiting effect.

[0057] FIG. 1A shows a block diagram of a feedback controller 110 for controlling the operation of a mechanical system subject to uncertainty on the state of the operation of the system according to some embodiments. The feedback controller 110 includes at least one processor and a non-transitory memory. The processor may be a single-core processor, a multi-core processor, a computing cluster, or any number of other configurations. The memory may include random access memory (RAM), read-only memory (ROM), flash memory, or any other suitable memory system. Additionally, in some embodiments, the memory may be implemented using a hard drive, an optical drive, a thumb

drive, an array of drives, or any combination thereof. Memory has instructions stored thereon that, when executed by the processor, cause the feedback controller to perform the control operations according to the embodiments.

[0058] To control the system, controller 110 collects 101 a feedback signal indicative of the current state of the operation of the system subject to constraints and current uncertainty on the current state of the operation of the system. Examples of constraints on the system include staying within an operating region, eventually reaching an operating region, avoiding obstacles in the environment. Exemplary uncertainties acting on the system include actuation uncertainty, errors in modeling, and sensing errors.

[0059] In a number of control applications employed by the embodiments, the current uncertainty lies in a symmetric bounded set and includes one or a combination of uncertainty on the dynamics of the system, uncertainty on a state of the system, uncertainty on control commands to the system, and uncertainty on constraints on the state of the operation of the system. For example, the symmetric bounded set containing the uncertainty may be an ellipsoid that has a pre-specified probability mass, and the ellipsoid is characterized either by the first two statistical moments of the uncertainty or under the assumption that the uncertainty has a Gaussian distribution. Another example for the symmetric bounded set is that the uncertainty lies in an axis-aligned hypercuboid where each edges of the hypercuboid indicates the uncertainty along each state dimension and the uncertainty is independent across the state dimensions.

[0060] Given the current state, constraints, and uncertainties of the operation, controller 110 determines 102 a robust controllable (RC) set for maintaining the state of the system within a prediction horizon. In various embodiments, the controller determines the RC set using closed-form expressions on a constrained zonotope modeling sets that characterize the constraints on the state of the operation of the system and an affine transformation of the symmetric bounded set inclosing the current uncertainty into a high-dimensional space of the constrained zonotopic representation. The closed-form expressions include a closed-form approximation of a Pontryagin difference between the constrained zonotopic representation and the zonotopic transformation of the symmetric bounded set allowing to determine the RC set in a computationally efficient manner.

[0061] Given the RC set, the controller determines 103 a control command for controlling the operation of the system, and submits the control command to an actuator of the system causing a change in the state of the operation of the system.

[0062] FIG. 1B shows a schematic of some principles employed by different embodiments. Some embodiments are based on the recognition that robust controllable sets can be computed using polyhedral computational geometry 110b. Specifically, the robust controllable sets can be obtained via a set recursion involving set operations like intersection, affine transformation, Minkowski sum, and Pontryagin difference. However, given one or more bounded polyhedra characterized by linear constraints, the Minkowski sum 120b operation requires solving a vertex-facet enumeration problem, which is known to be NP-hard. Consequently, existing approaches using polyhedral computational geometry suffer from numerical issues when used in high dimensions, and are limited to low-dimensional systems.

[0063] Some embodiments are based on recognizing that constrained zonotopes **130b** provide an alternative representation of bounded polyhedrons (also called polytopes). A constrained zonotopic representation of a polytope characterizes the polytope as an affine transformation (projection) of the intersection of a unit hyper-cube in a high-dimensional space and an affine set. Affine sets are solution sets to some collection of linear equations. Constrained zonotopes provide closed-form, easy-to-evaluate expressions to compute the intersection, affine transformation, and Minkowski sum. However, no closed-form expression exists for the computation of the Pontryagin difference **140b**.

[0064] Some embodiments are based on recognizing that instead of considering the generic form of the Pontryagin difference, a computationally tractable approach for the Pontryagin difference between a constrained zonotope and a centrally symmetric, affine transformation, and compact set is what is needed for many practical control applications. In addition, for these types of control applications, the Pontryagin difference can be inner- or outer-approximated with closed-form expressions **150b**.

[0065] FIG. 1C shows an illustration of the benefits of the proposed method in computing robust controllable sets using constrained zonotopes computational geometry **130b**, according to an embodiment of the present disclosure. Schematic **105** shows an illustration of a robust controllable set **108** computed for the set of constraints **107** based on some known post-failure characteristics. Boxes **106** and **110** highlight an advantage and three main shortfalls of the existing approach (shown in box **106**) over the proposed method (shown in box **110**). Specifically, the existing approach of using polytopes to compute the robust controllable set has the advantage of being applicable to a broad range of bounded uncertainties. On the other hand, the proposed approach requires the uncertainty to be symmetric about a point, which encompasses a sufficiently large class of uncertainty models. However, the proposed approach has three main advantages over the existing approaches. First, the proposed method admits closed-form expressions for all the necessary set operations used in robust controllable set computation. Consequently, the proposed method is orders of magnitude faster than the existing approaches for large-dimensional systems or for set computation over long horizons. Second, the proposed method's speed allows for possible online computation of the sets, which can not be accomplished with existing approaches. Third, the proposed method uses standard linear algebra to arrive at the closed-form expressions and does not require a specialized tool to solve the vertex-facet enumeration problem that plagues the existing approach.

Polytopes, Zonotopes, and Constrained Zonotopes

[0066] Different embodiments use different properties of computational geometry employing a constrained zonotope.

[0067] FIG. 2A illustrates a convex and compact set **201** that is centrally symmetric about point **202** having properties employed by some embodiments. Set **201** is convex since the line segment **205** joining any two points **203** and **204** is completely contained in set **201**. In Euclidean spaces, the set **201** is compact since it is closed and bounded. Finally, set **201** is centrally symmetric about point **202**, since for any point **206** in set **201**, its reflection **207** about point **202** is also contained in set **201**.

[0068] As used herein, polytopes are a collection of linear constraints that together form a bounded set. A polytope is formally defined as

$$\mathcal{P} = \{x \mid a_i^T x \leq b_i \text{ for } i = 1, 2, \dots, N_P\} \subset \mathbb{R}^n. \quad (1)$$

[0069] Here, $a_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$ are vectors and scalars that define the individual linear constraints that together (upon their intersection) form $\mathcal{P}$. Polytopes form a broad class of convex and compact sets, and it is well-known that any arbitrary convex and compact set can be approximated to a desired accuracy using a polytope. The polytope described in (1) is said to be in its half-space representation form. Alternatively, the polytope can be described as a convex combination of its vertices, in which case it is said to be in its vertex representation form. Recall that converting a polytope from one form to another is known as the vertex-facet enumeration problem, and is known to be NP-hard.

[0070] FIG. 2B illustrates a zonotope employed by some embodiments. Zonotopes are a special class of polytopes that are defined by an affine transformation of a unit box. Specifically, set **211** is a zonotope that was defined by an affine transformation **215** of a unit box **212**. Observe how the affine transformation **215** deforms the standard axes **213** that are normal to the sides of the unit box **212** to define a new set of deformed axes **218** that are normal to the sides of the zonotope. Formally, a zonotope is defined as

$$\mathcal{Z} = \{G\xi + c \mid \|\xi\|_\infty \leq 1\} \subset \mathbb{R}^n. \quad (2)$$

[0071] Here, $G \in \mathbb{R}^{n \times N_Z}$ and $c \in \mathbb{R}^n$ are a matrix and a vector respectively that together define the affine transformation **215**. The unit box **212** is defined by the set $\{\|\xi\|_\infty \leq 1\} \subset \mathbb{R}^{N_Z}$. While FIG. 2B shows the case where $n=N_Z$, it need not always be the case. In the case where $n < N_Z$, the affine transformation projects the higher-dimensional unit box **212** to a lower-dimensional set **211**. By construction, a zonotope is a convex, compact set that is centrally symmetric about c **219**.

[0072] Constrained zonotopes are closely related to polytopes (1) and zonotopes (2). Formally, a constrained zonotope is defined as

$$\mathcal{C} = \{G_C \xi + c_C \mid \|\xi\|_\infty \leq 1, A_C \xi = b_C\} \subset \mathbb{R}^n. \quad (3)$$

[0073] Similarly to (2), $G_C \in \mathbb{R}^{n \times N_C}$ and $c_C \in \mathbb{R}^n$ are a matrix and a vector respectively that together define the affine transformation of the high-dimensional space $\mathbb{R}^{N_C}$ (since $N_C > n$ typically). However, unlike (2), (3) includes a collection of linear equality constraints $A_C \xi = b_C$.

[0074] FIG. 2C illustrates a constrained zonotope employed by some embodiments. The set **221** is a constrained zonotope. It is defined in two stages. Similarly to FIG. 2B, the unit box **222** is defined by the set $\{\|\xi\|_\infty \leq 1\} \subset \mathbb{R}^{N_C}$. In the first stage, a unit box **222** is intersected with a collection of linear constraints $\{\xi \mid A_C \xi = b_C\} \subset \mathbb{R}^{N_C}$ that form a hyperline **223**. The intersection operation **225** produces the set **224**. The set **224** is formally defined as $\{\xi \mid \|\xi\|_\infty \leq 1,$

$A_C \xi = b_C\} \subset \mathbb{R}^{N_C}$. Observe that the coordinate space **227** has not changed under this operation. Next, in the second stage, the set **224** undergoes an affine transformation **229** as characterized by (G_C, c_C) . In this case, observe that the three-dimensional set **224** ($N_C=3$) was projected to a two-dimensional set **221** ($n=2$), as indicated by the coordinate axes **228**.

[0075] Constrained zonotopes are an alternative description of a polytope and some embodiments benefit from understanding that every polytope (1) has a constrained zonotopic representation, and vice-versa.

Set Operations

[0076] For any sets $\mathcal{C}, \mathcal{S} \subseteq \mathbb{R}^n$ and $\mathcal{W} \subseteq \mathbb{R}^m$, and a matrix $R \in \mathbb{R}^{m \times n}$, define the set operations (affine map, Minkowski sum $\oplus$, intersection with inverse affine map $\cap_R$, and Pontryagin difference $\ominus$):

$$\mathcal{RC} \triangleq \{Ru : u \in \mathcal{C}\}, \quad (4)$$

$$\mathcal{C} \oplus \mathcal{S} \triangleq \{u + v : u \in \mathcal{C}, v \in \mathcal{S}\}, \quad (5)$$

$$\mathcal{C} \cap_R \mathcal{W} \triangleq \{u \in \mathcal{C} : Ru \in \mathcal{W}\}, \quad (6)$$

$$\mathcal{C} \ominus \mathcal{S} \triangleq \{u : \forall v \in \mathcal{S}, u + v \in \mathcal{C}\}. \quad (7)$$

[0077] For polytopes $\mathcal{C}, \mathcal{S} \subseteq \mathbb{R}^n$ and $\mathcal{W} \subseteq \mathbb{R}^m$, all the above operations yield polytopes. However, their computation is complicated by the fact that the exact implementation of operations (4) and (5) require the polytopes in vertex representation form (especially in higher dimensions), while operations (6) and (7) require the polytopes in half-space representation form. Furthermore, the process of converting a polytope from its half-space representation form to its vertex representation form is known to be an NP-hard problem.

[0078] Some embodiments are based on the recognition that there is a need for tractable approaches to implement the set operations (4), (5), (6), and (7) to compute robust controllable sets and stochastic controllable sets.

[0079] Some embodiments are based on the recognition that constrained zonotopes have closed-form expressions for the above set operations, except for the Pontryagin difference:

$$\mathcal{RC} = (R\mathcal{C}, \mathcal{RC}_C, A_C, b_C), \quad (8)$$

$$\mathcal{C} \oplus \mathcal{S} = ([G_C, G_S], c_C + c_S, [A_C, 0; 0, A_S], [\mathcal{C}; b_S]), \quad (9)$$

$$\mathcal{C} \cap_R \mathcal{W} = \quad (10)$$

$$([G_C, 0], c_C, [A_C, 0; 0, A_W \mathcal{R} G_C, -G_W], [\mathcal{C}; b_W \mathcal{C}_W - \mathcal{RC}_C]),$$

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[0080] It is an object of some embodiments to provide a method to compute an inner- and outer-approximation of the Pontryagin difference of a constrained zonotope and a convex, compact set that is symmetric about some point.

Pontryagin Difference Using Constrained Zonotopes

[0081] FIG. 3A illustrates the Pontryagin difference between two sets employed by some embodiments. Specifi-

cally, compute $\mathcal{C} \ominus \mathcal{S}$ which is the set **301**, where $\mathcal{C}$ is a rectangle **302** and $\mathcal{S}$ is a circle **303**. The operation **305** implements (7). Hence, the set **301** satisfies the requirement that for any $x \in \mathcal{C} \ominus \mathcal{S}$, the set $\mathcal{S}$ translated by x is contained in $\mathcal{C}$. This is evidenced by the sets **303**, **304**, **305**, and **306**. [0082] FIG. 3B illustrates an inner-approximation and outer-approximation of the set **301** employed by some embodiments. It is an object of some embodiments to provide a method to compute an inner-approximation **308** and outer-approximation **309** of the Pontryagin difference **301** when the set $\mathcal{C}$ is a constrained zonotope and the set $\mathcal{S}$ is a convex, compact set that is symmetric about some point.

[0083] FIG. 3C shows various examples of sets that are convex, compact, and symmetric about some point having principles employed by some embodiments. The set **211**, which is a zonotope as defined by (2), is a convex and compact set that is symmetric about the point c **219**, as shown in FIG. 2B. Some other examples include ellipsoids and convex unions of symmetric intervals. Formally, these sets are given as follows:

$$\exists (G_Z, c_Z) \in \mathcal{Z} \times \mathbb{R}^n : \mathcal{Z} = \{G_Z \xi + c_Z : \|\xi\|_\infty \leq 1\} \quad (11)$$

$$\exists (G_E, c_E) \in \mathcal{E} \times \mathbb{R}^n : \mathcal{E} = \{G_E \xi + c_E : \|\xi\|_2 \leq 1\}, \quad (12)$$

$$\exists (\mathcal{I}) \in \mathcal{I} \times \mathbb{R}^n : \mathcal{I} = \{\mathcal{I} \xi + \mathcal{I} : \|\xi\|_1 \leq 1\}, \quad (13)$$

Ⓜ indicates text missing or illegible when filed

[0084] Here, (11) is exactly the same as (2), while (12) defines an ellipsoid **310** symmetric about a point **311**, and (13) defines a convex union of symmetric intervals **315** symmetric about a point **316**. Both the sets **310** and **315** are convex and compact. Observe that the set **315** is a convex hull of the union of intervals are of the form $[-g_i, g_i]$ where g_i are the columns of the matrix G_I . Specifically, it is the convex hull of the union of the intervals **327**, **328**, and **329** that is formed by **322** and **325**, **323** and **326**, and **321** and **324**.

[0085] Some embodiments are based on the recognition that it is possible and feasible to compute a constrained zonotope $\mathcal{M}^-$ that inner-approximates $\mathcal{M} = \mathcal{C} \ominus \mathcal{S}$, given a non-empty constrained zonotopic minuend $\mathcal{C} = (G_C, c_C, A_C, b_C) \subset \mathbb{R}^n$ and a convex and compact subtrahend $\mathcal{S} \subset \mathbb{R}^n$ that is symmetric about any $c_S \in \mathbb{R}^n$. Specifically, let $\Gamma : \mathbb{R}^{N_C \times n}$ solve the following system of linear equations,

$$\begin{bmatrix} G_C \\ A_C \end{bmatrix} \Gamma = \begin{bmatrix} I_n \\ \mathcal{I} \end{bmatrix}, \quad (14)$$

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$D : \mathbb{R}^{N_C \times N_C} D_{ii} = 1 - \rho_S(e_i^T \Gamma) \rho_S(\ell) = (\max_{x \in \mathcal{S}} \ell^T x) + \ell^T c_S e_i$
 $\mathbb{R}^{N_C \times 1} = 1, 2, \dots, N_C \mathcal{M}^- = (G_C D, c_C - c_S, A_C D, b_C) D_{ii} \geq 0 i = 1, 2, \dots, N_C$ and let be a diagonal matrix with diagonal entries
 $D : \mathbb{R}^{N_C \times N_C} D_{ii} = 1 - \rho_{-c_S}(e_i^T \Gamma) \rho_{-c_S}(\ell) = (\max_{x \in \mathcal{S}} \ell^T x) + \ell^T c_S e_i$
 $\mathbb{R}^{N_C \times 1} = 1, 2, \dots, N_C \mathcal{M}^- = (G_C D, c_C - c_S, A_C D, b_C) D_{ii} \geq 0 i = 1, 2, \dots, N_C$, where ρ denotes the standard axes in for each. Then, the constrained zonotope, provided for every.

[0086] FIG. 3D shows pseudocode of an algorithm of a method for computing an inner-approximation of the Pontryagin difference between a constrained zonotope $\mathcal{C}$ and a convex, compact set $\mathcal{S}$ that is symmetric about a point c_s according to some embodiments. In Algorithm 1, Line 1 computes a set $\mathcal{S}_0$ corresponding to that translates $\mathcal{S}$ by the point of symmetry $-c_s$, such that $\mathcal{S}_0$ is now convex and compact, and symmetric about the origin. Next, in Line 2, computes a solution to (14) using the Moore-Penrose pseudoinverse of the coefficient matrix in (14). Next, in Line 3, it sets up a diagonal matrix D and, in Line 5, returns the inner-approximation to the Pontryagin difference $\mathcal{C} \ominus \mathcal{S}$, provided all the diagonal entries of D are non-negative.

[0087] FIG. 3E illustrates the intuition behind the algorithm described in FIG. 3D used to compute an inner-approximation of the Pontryagin difference between a constrained zonotope and a convex, compact set that is symmetric about a point. Specifically, it computes an inner-approximation $\mathbf{331}$ of the Pontryagin difference $\mathbf{330}$ between the constrained zonotope $\mathbf{211}$ as defined in FIG. 2C and the convex union of symmetric intervals as defined in FIG. 3C. After solving the linear equations (14), we define the set $\mathcal{V}_0 = \Gamma(\mathcal{S} - c_s)$. The set $\mathcal{V}_0$ is $\mathbf{332}$ obtained via an affine transformation $\mathbf{333}$ of $\mathcal{S}$, the set $\mathbf{315}$. Observe that set $\mathbf{332}$ is often of a higher dimension than set $\mathbf{315}$, but is of the same dimension as set $\mathbf{224}$. Next, we can show that the Pontryagin difference (denoted by operation $\mathbf{335}$) between sets $\mathbf{224}$ and $\mathbf{332}$, denoted by set $\mathbf{334}$, is available in closed-form. Consequently, using the affine transformation $\mathbf{229}$, we obtain an inner-approximation inner-approximation $\mathbf{331}$ of the Pontryagin difference $\mathbf{330}$.

[0088] FIG. 4A illustrates an algorithm to compute the polytopic outer-approximation of a constrained zonotope according to some embodiments. As discussed in FIG. 3B., Algorithm 2 described in FIG. 4A provides a convex polyhedron $\mathcal{P}$ that contains a given full-dimensional constrained zonotope $\mathcal{C}$. In Algorithm 2, Line 1 sets up a matrix V where the columns v_i solve an equation similar to (14), and normalize it such that each of the columns v_i lie in a unit ℓ_1 -norm ball. Next, Lines 2 and 3 use simple affine transformations to generate the half-space representation of the polyhedron that outer-approximates $\mathcal{C}$.

[0089] FIG. 4B shows pseudocode of an algorithm of a method for computing a constrained zonotopic outer-approximation of the Pontryagin difference between a constrained zonotope and a convex, compact set that is symmetric about a point according to some embodiments. The algorithm uses the observation that the Pontryagin difference between a polyhedron and a convex and compact set is available in closed form $\mathcal{M}_p^+$ (Line 2), and that the constrained zonotope $\mathcal{C} - c_s$ is guaranteed to contain the set $\mathcal{M} = \mathcal{C} \ominus \mathcal{S}$. Consequently, the last step of Algorithm 3 produces a constrained zonotope $\mathcal{M}^+$ that is guaranteed to contain the set $\mathcal{M} = \mathcal{C} \ominus \mathcal{S}$ (Line 3).

Exemplary Embodiment of Fault-Tolerant Control: Abort-Safe Spacecraft Control Under Uncertainty

[0090] FIG. 5A illustrates the motion of spacecraft $\mathbf{591}$ in a multi-object celestial system, according to an embodiment of the present disclosure. The spacecraft $\mathbf{591}$ may be configured to rendezvous with a target $\mathbf{593}$ by following a trajectory $\mathbf{594}$. The target $\mathbf{593}$ may be a spacecraft, a celestial body, an international space station, or orbital debris. For the

purpose of explanation, the target is $\mathbf{593}$ is illustrated as the International Space Station. A keep-away zone $\mathbf{592}$ exists around target $\mathbf{593}$. The keep-away zone $\mathbf{592}$ refers to a region that spacecraft $\mathbf{591}$ should not enter without authorization. Additionally or alternatively, the keep-away zone $\mathbf{592}$ corresponds to a keep-away set, which is a collection of states including spacecraft positions and velocities that spacecraft $\mathbf{591}$ should stay outside of.

[0091] It is an objective of some embodiments to control the motion of spacecraft $\mathbf{591}$ while avoiding an unauthorized entry into the keep-away zone $\mathbf{592}$ during a normal and abnormal (post-failure) operation of spacecraft $\mathbf{591}$. As used herein, the normal operation includes spacecraft $\mathbf{591}$ moving towards the keep-away zone $\mathbf{592}$. The abnormal (post-failure) operation includes one or a combination of a failure to receive the authorization to enter the keep-away zone $\mathbf{592}$ and a failure of at least one component of the spacecraft $\mathbf{591}$, for example, thrusters.

[0092] To achieve such a control objective, some embodiments provide a controller $\mathbf{110}$ for controlling the motion of spacecraft $\mathbf{591}$ during the normal and abnormal (post-failure) operation of spacecraft $\mathbf{591}$.

[0093] Some embodiments are based on the recognition that the post-failure desirable behavior may be enforced on the nominal trajectory via a collection of constraints that guarantee post-failure safety. Moreover, some embodiments are based on the recognition that these constraints may be computed using robust or stochastic controllable sets based on the post-failure characteristics.

[0094] Some embodiments are based on a recognition that achieving the aforementioned control objective (i.e., controlling the motion of spacecraft $\mathbf{591}$ while avoiding an unauthorized entry into the keep-away zone $\mathbf{592}$ during a normal and abnormal (post-failure) operation of spacecraft $\mathbf{591}$ is challenging because there are states of the spacecraft $\mathbf{591}$ in proximity of the keep-away zone $\mathbf{592}$ that may unavoidably lead the spacecraft $\mathbf{591}$ into the keep-away zone $\mathbf{592}$, even if the authorization to enter is rejected. The authorization to enter may be rejected based on any reason including a total or partial failure of propulsion of the spacecraft $\mathbf{591}$. As used herein, the state of spacecraft $\mathbf{591}$ includes the location of spacecraft $\mathbf{591}$ and at least one or a combination of a velocity and an acceleration of spacecraft $\mathbf{591}$. Additionally, internal and external forces acting on spacecraft $\mathbf{591}$ during its motion, such as inertia and force of gravity, may change the state of spacecraft $\mathbf{591}$ to specific states that may lead the spacecraft $\mathbf{591}$ into the keep-away zone $\mathbf{592}$, regardless of any efforts that the spacecraft $\mathbf{591}$ may perform.

[0095] Some embodiments are based on a recognition that the motion of the spacecraft $\mathbf{591}$ may be subject to uncertainty. The process noise may specify an actuation mismatch and/or unmodeled dynamics of the motion of the spacecraft $\mathbf{591}$. In addition to the process noise, the stochastic uncertainty may include measurement noise of an estimation of the state of the spacecraft. Since the motion of spacecraft $\mathbf{591}$ is subject to uncertainty, there is a need to avoid the unauthorized entry of spacecraft $\mathbf{591}$ into the keep-away zone $\mathbf{592}$ despite the uncertainty.

[0096] Some embodiments are based on the realization that this problem can be addressed by computing a union of a plurality of robust/stochastic controllable sets of values of the state of spacecraft $\mathbf{591}$ that the state of spacecraft $\mathbf{591}$ should be maintained within. The choice between the use of

robust or stochastic controllable sets depends on the information available about the uncertainty in the problem.

[0097] When the uncertainty (process and measurement noise) is known to lie in pre-defined sets but can take any value in the set, then one can use robust controllable sets. Specifically, by ensuring that the designed rendezvous trajectory belongs to appropriate robust controllable sets, the trajectory is guaranteed to be abort-safe irrespective of the value taken by the uncertainty. However, depending upon the size of the pre-defined sets modeling the uncertainties, the resulting trajectory can be quite conservative (they may take more fuel or stop further away from the keep-out set) to hedge against the worst possible (in terms of the safety requirements) realization of the uncertainty.

[0098] On the other hand, if a probabilistic description of the uncertainty is made available or if the designer has access to sets in which the uncertainty is guaranteed to lie in a designer-specified likelihood, then one can use stochastic controllable sets and chance constraints. Specifically, by ensuring that the designed rendezvous trajectory belongs to appropriate stochastic controllable sets with appropriately chosen likelihoods (using chance constraints), the trajectory is guaranteed to be abort-safe with high likelihood. Compared to robust controllable sets, the trajectories designed using stochastic controllable sets are typically less conservative (they may take less fuel or stop closer to the keep-out set) since the problem formulation now allows for a small likelihood of unsafe behavior.

[0099] FIG. 5B shows a block diagram of a trajectory design method 120 for fault-tolerant applications that use set-based control according to some embodiments. Method 120, starting with the current state 125 of the system, designs a control action to apply 126 to spacecraft 591 based on whether a failure 127 has occurred or not.

[0100] The inputs to the method 120 are highlighted in dashed boxes - - the current state of the system 125, characteristics of the system under nominal operation 135 like the nominal dynamics 132 and the nominal uncertainty model 133, constraints to guarantee nominal task satisfaction 141, objective for nominal control design 145, characteristics of the system under post-failure operation 151 like the post-failure dynamics 153, the post-failure uncertainty model 154, and the post-failure safety constraints 152.

[0101] The output of method 120 is a control action 126 that depends on the current state of system 125 and whether a failure 127 has occurred or not. In the spacecraft rendezvous, an example of failure may be that a thruster has failed and now the spacecraft only has limited actuation and is now subject to dynamics 153 and uncertainty model 154 which is different from nominal dynamics 132 and nominal uncertainty model 133.

[0102] When a failure has not occurred, method 120 designs the control to apply for nominal operation 130 using a control law for nominal operation 131. Such a control law can be designed using existing control paradigms like model predictive control or constrained control theory. The control law in 131 enforces the characteristics of the system under nominal operation 135 like the nominal dynamics 132 and the nominal uncertainty model 133.

[0103] Additionally, control law 131 ensures that the constraints to guarantee the completion of tasks in nominal operation 142 are enforced. In the spacecraft rendezvous, an example of such a constraint 142 is that the nominal rendezvous must lie inside a line-of-sight cone originating from

the target station. Such constraints are hard constraints and may not be violated during the nominal operation of the system. In some implementations, method 120 uses constraint tightening 141 to arrive at a set of tightened constraints to guarantee nominal task constraints 140, whose enforcement ensures that the use of control law 131 will ensure that the nominal system 132 satisfies constraints 142, despite the nominal uncertainty 133. Additionally, it optimizes the objective for nominal control design that is optimized during the design of the control law 131. The objective is typically composed of one or more performance objectives and soft constraints. In the spacecraft rendezvous, an example of the performance objective may be the energy used during the nominal maneuver, and an example of soft constraints may be that the spacecraft during the nominal maneuver should point towards a star. Soft constraints, unlike the hard constraints specified in 142, may be violated if a need arises.

[0104] Additionally, the control law in 131 must ensure that the system is always maintained in a configuration to guarantee post-failure safety in the event of a failure. Specifically, the system must still satisfy post-failure desirable behavior encoded in post-failure safety constraints 152 under post-failure dynamics 153 and post-failure uncertainty model 154. In the spacecraft rendezvous, an example of the post-failure safety may be that, in the event of a thruster failure, the spacecraft can still avoid the target under limited actuation (potentially via the reaction control systems) and a dynamics model that may be different from the nominal dynamics due to the thruster failure.

[0105] Some embodiments are based on the recognition that the post-failure desirable behavior encoded in 151 may be enforced on the nominal trajectory via a collection of constraints 157 that guarantee post-failure safety. Moreover, some embodiments are based on the recognition that these constraints 157 may be computed using robust or stochastic controllable sets based on the post-failure characteristics 151. These controllable sets also provide a control law for post-failure operation 159.

[0106] Similarly to steps 142 and 140, some embodiments use constraint tightening 158 on the identified constraints 157 to obtain a set of tightened constraints on the nominal trajectory that guarantees post-failure safety.

[0107] Thus, the control law for the nominal operation 131 takes as input the current state 125, the tightened constraints to guarantee nominal task satisfaction 140, the objective for nominal control design 145, the characteristics of the system under nominal operation 135 including the nominal dynamics 132 and the nominal uncertainty model 133, and the tightened constraints to guarantee post-failure safety 150, and provides a control action 130 to apply under nominal operation.

[0108] On the other hand, when a failure has occurred, the embodiments use the control law for post-failure operation 159 and the current state 125 to obtain a control action 160 to apply.

[0109] The control that is finally applied to system 126 is selected from the control action to apply under nominal operation 130 and the control action to apply in the event of failure 160, based on whether a failure has occurred 127.

[0110] It is an object of some embodiments to provide a system and a control system and a method to design the constraints 157 to guarantee post-failure safety using robust controllable sets or stochastic controllable sets that are

approximated using constrained zonotopes. The proposed approach has a significant computational advantage over existing approaches without compromising the accuracy of the computed sets.

[0111] In some embodiments, different implementations consider two types of post-failure uncertainty models in post-failure uncertainty model 154, i.e., bounded non-stochastic uncertainty and stochastic uncertainty. In both cases, the embodiments use reachability to identify the constraints to guarantee post-failure safety 157.

[0112] In such a manner, some embodiments disclose a spacecraft 591 for moving in a multi-object celestial system while avoiding an unauthorized entry into a keep-away zone during a normal and an abnormal operation of the spacecraft, wherein the normal operation includes moving towards a target in the keep-away zone, and wherein the abnormal operation includes one or a combination of a failure to receive an authorization to enter the keep-away zone and a failure of at least one component of the spacecraft.

[0113] The spacecraft includes the feedback controller 110, a set of thrusters configured to change the state of a spacecraft according to a sequence of control commands produced by the feedback controller 110, a set of sensors configured to produce measurements indicative of the state of the spacecraft; and a circuitry configured to detect the abnormal operation of the spacecraft.

[0114] In this embodiment, the feedback controller is a fault-tolerant controller is configured to perform an abort-safe control under the uncertainty. To that end, the feedback controller is configured to execute, during the normal operation of the spacecraft, a nominal control law subject to constraints on maintaining a state of the spacecraft within a union of a plurality of control invariant sets of values of the state of the spacecraft that partially or completely enclose the keep-away zone, wherein the state of the spacecraft includes a location of the spacecraft and at least one or a combination of a velocity and an acceleration of the spacecraft, wherein each of the plurality of control invariant sets is determined using constrained zonotopes computational geometry such that when the state of the spacecraft is within a control invariant set there is a control command produced by the nominal control law that maintains the state of the spacecraft within the control invariant set despite internal and external forces acting on the spacecraft; and execute, upon detecting the abnormal operation of the spacecraft, an abort control law associated with the control invariant set including a current state of the spacecraft, wherein at least some different abort control laws are associated with at least some different control invariant sets, and wherein the abort control law is jointly and interdependently determined for the corresponding control invariant set to produce abort control commands moving the spacecraft while avoiding the keep-away zone for any state within the corresponding control invariant set.

Case 1: Robust Controllable Sets Via Constrained Zonotopes (Post-Failure Uncertainty is a Bounded Non-Stochastic Uncertainty)

[0115] In some implementations, each of the control invariant sets is a stochastic reachable set determined for a possible abnormal operation defined by the first likelihood of the unbounded stochastic uncertainty.

[0116] Robust controllable (RC) sets, also known as robust backward reach sets, characterize a set of states from

which a collection of possibly time-varying state constraints can be satisfied by a controlled state trajectory, under bounded control authority and bounded uncertainty. RC sets are essential for robust model predictive control, fault-tolerant control, and verification of dynamical systems, and have been used in a broad range of applications in space, transportation, and robotics. For discrete-time linear systems with additive uncertainty and bounded, polytopic state and input constraints, exact RC sets are known to be polytopes, and the RC sets may be computed using set manipulations of polytopes. Polytopes are convex geometric shapes, and their manipulations have been well-studied in the literature. However, polytope-based RC set computation involves a projection that either can be accomplished via vertex-facet enumeration or Fourier-Motzkin elimination or multi-parametric optimization. These operations are either computationally prohibitive or cause numerical issues for high-dimensional system models arising from complex systems and/or in applications involving long safety horizons.

[0117] FIG. 5C shows an example of a two-dimensional projection 500 of a robust controllable set 503 corresponding to a constraint set 501, according to an embodiment of the present disclosure. In an embodiment, the constraint set 501 may be a multi-dimensional polytope determined by hyperplanes, which are represented by linear inequalities, along multiple dimensions corresponding to the constraints on the motion of the spacecraft 591. The constraint set 501 may encode the states of the spacecraft 591 that are safe. Any state of the spacecraft 591 within the robust controllable set 503, there exists a control command maintaining the state of the spacecraft 591 within the constraint set 501 for the known or the admissible future states of the nominal trajectory, despite the bounded uncertainty affecting the nominal dynamics. For example, for any state of the spacecraft 591 such as a state 515 within the robust controllable subset 503 and within all possible control inputs 517-523 that the controller 110 can execute, there is at least one control command 523 that maintains the state of the spacecraft 591 within the robust controllable subset 503, despite the bounded uncertainty affecting the nominal dynamics. On the other hand, a state 505 can be feasible for one iteration, but all control commands 507-513 that the controller 110 is allowed to take during the next iteration can bring the state of the spacecraft 591 outside of the constraint set 501, possibly due to the bounded uncertainty affecting the nominal dynamics.

[0118] Consider a linear time-varying system with the following dynamics

$$x_{t+1} = A_t x_t + B_t u_t + F_t w_t \quad (15)$$

with state $x_t \in \mathbb{R}^n$, input $u_t \in \mathcal{U}_t \subset \mathbb{R}^m$, disturbance $w_t \in \mathcal{W}_t \subset \mathbb{R}^p$, where the (possibly time-varying) sets $\mathcal{U}_t$, $\mathcal{W}_t$ are assumed to be convex and compact, and the (possibly time-varying) matrices A_t , B_t , and F_t are appropriately defined for $t=0, 1, \dots$. For some time horizon $T \in \mathbb{N}$, let X_t denote the (possibly time-varying) collection of constraints for $t=0, 1, \dots, T$.

[0119] Then, we compute the robust controllable set $\mathcal{K} = \mathcal{K}_0$ using the following set recursion with $\mathcal{K}_T = X_T$:

$$\textcircled{2} = \text{Pre}(\mathcal{K}_{t+1}) \cap X_t, \forall t = 0, 1, \dots, T-1, \quad (16a)$$

$$\text{Pre}(\mathcal{K}_{t+1}) \triangleq \{x: A_t x \in (\mathcal{K}_{t+1} \ominus F_t W_t) \oplus (-B_t U_t)\}, \quad (16b)$$

② indicates text missing or illegible when filed

[0120] Observe that the set recursion (16) uses all the set operations defined in (4)-(7). The sets $\mathcal{K}_t$ is guaranteed to be convex and compact. Moreover, these sets are guaranteed to be polytopic if the time-varying sets involved U_t, X_t are also polytopes. However, the computation of $\mathcal{K}_t$ using polyhedral computational geometry is complicated by the fact that implementation of all set operations (4)-(7) requires the NP-hard problem of vertex-facet enumeration problem.

[0121] Some embodiments are based on the recognition that the set operations in (16) can be implemented exactly or approximated using constrained zonotopes without solving the NP-hard problem of vertex-facet enumeration.

[0122] In the following, we will assume that the input set U_t and the terminal constraint set X_T are convex polytopes, hence, representable as constrained zonotopes, and the additive disturbance set W_t is a convex and compact set that is symmetric about any $c_W \in \mathbb{R}^P$. We will also use the fact that converting a convex polytope in half-space representation to an equivalent set in the constrained zonotopic is easily accomplished using optimization. Consequently, any convex polytope may also be thought of as a constrained zonotope.

[0123] Specifically, for a convex polyhedral X_t and invertible system matrix A_t , we can use the following set recursion:

$$\mathcal{K}_{t,inner}^{interim,1} \subseteq \mathcal{K}_t^{interim,1} = \mathcal{K}_{t+1} \ominus F_t W_t, \quad (17)$$

$$\mathcal{K}_t^{interim,2} = \mathcal{K}_{t,inner}^{interim,1} \oplus (-B_t U_t), \quad (18)$$

$$\mathcal{K}_t^{interim,3} = A_t^{-1} \mathcal{K}_t^{interim,2}, \quad (19)$$

$$\mathcal{K}_T = X_T W_t, \mathcal{K}_t = \mathcal{K}_t^{interim,3} \cap X_t, \quad (20)$$

$$\mathcal{K}_T = X_T W_t$$

$\mathcal{K}_T = X_T W_t$ where the recursion is initialized with (a convex polytope, and hence representable via a constrained zonotope). Note that the steps (18)-(20) have closed-form expressions for the constrained zonotopic sets (see (8) to (10)) and the fact that constrained zonotope intersected with a halfspace also has a closed-form expression (see Algorithm 3 in FIG. 4B). Finally, the step (17) has an inner-approximation in the form of a constrained zonotope using Algorithm 1 in FIG. 3D and an outer-approximation in the form of a constrained zonotope using Algorithm 2 in FIG. 4B. Note that both algorithms have closed-form expressions when the disturbance set is an ellipsoid, zonotope, or a convex union of symmetric intervals.

[0124] Alternatively, for a convex polytopic X_t and (possibly non-invertible) system matrix A_t , we can use the following set recursion:

$$\mathcal{K}_{t,inner}^{interim,1} \subseteq \mathcal{K}_t^{interim,1} = \mathcal{K}_{t+1} \ominus F_t W_t, \quad (21)$$

$$\mathcal{K}_t^{interim,2} = \mathcal{K}_{t,inner}^{interim,1} \oplus (-B_t U_t), \quad (22)$$

$$\mathcal{K}_t = X_t \cap_{A_t} \mathcal{K}_t^{interim,2}, \quad (23)$$

where the recursion is initialized with $\mathcal{K}_T = X_T$ (a convex polytope, and hence representable via a constrained zonotope). Note that steps (22) and (23) have closed-form expressions for the constrained zonotopic sets (see (8) to (10)). Finally, step (21) has an inner-approximation in the form of a constrained zonotope using Algorithm 1 in FIG. 3D and an outer-approximation in the form of a constrained zonotope using Algorithm 2 in FIG. 4B. Note that both algorithms have closed-form expressions when the disturbance set W_t is an ellipsoid, zonotope, or a convex union of symmetric intervals.

Case 2: Stochastic Controllable Sets Via Constrained Zonotopes (Post-Failure Uncertainty is a Stochastic Uncertainty)

[0125] In alternative implementations, each of the control invariant sets is a robust control invariant set determined for the nominal control law with a bounded non-stochastic uncertainty corresponding to the first likelihood on the unbounded stochastic uncertainty.

[0126] Stochastic controllable (SC) sets are defined similarly to RC sets. Specifically, SC sets characterize a set of states from which a collection of possibly time-varying state constraints can be satisfied with a user-specified likelihood by a controlled state trajectory, under bounded control authority and stochastic uncertainty. The differences between SC sets and RC sets are two folds. First, SC sets accommodates stochastic (possibly unbounded) uncertainty, whereas RC sets require the uncertainty to be bounded but does not require any stochastic information. Second, while RC sets guarantee absolute constraint satisfaction, SC sets provide constraint satisfaction up to a user-specified likelihood. Like RC sets, SC sets are essential for stochastic model predictive control, fault-tolerant control, and verification of dynamical systems, and have been used in a broad range of applications in space, transportation, and robotics.

[0127] It is well-known that SC sets can be inner-approximated using appropriately defined RC sets. Consequently, the existing approaches to inner-approximate SC sets suffer from the same challenges identified above, and it is a goal of this disclosure to remedy these shortcomings in computation of SC sets by computing an inner-approximation of the SC sets using constrained zonotopes.

[0128] Each stochastic controllable set used to determine the robust controllable set inner-approximates the values of the state of the spacecraft 591 to guarantee the control invariance. In an embodiment, the inner-approximation of the stochastic controllable set is obtained as a union of stochastic controllable sets defined for each convex component of a complement of the keep-out set. An inner approximation of the stochastic controllable set is described below with reference to FIGS. 6A, 6B, and 6C.

[0129] FIG. 6A shows a keep-away set 601 and its complement 606, according to an embodiment of the present disclosure.

[0130] FIG. 6B illustrates a stochastic controllable set **605**, according to an embodiment of the present disclosure. The stochastic controllable set **605** includes all states from which the controller **110** can maintain the spacecraft **591** safe (i.e., maintain the spacecraft **591** outside of the keep-away set **601**) up to a specified likelihood despite the unbounded stochastic uncertainty. For an initial state **607**, trajectories **611**, **616**, and **615** may be stochastic future state trajectories. For an initial state **609**, trajectories **617**, **619**, and **621** may be the stochastic future state trajectories. Since the spacecraft **591** is required to stay outside the keep-away set **601**, the trajectories **611**, **616**, and **621** are unsafe, while the trajectories **615**, **617**, and **619** are safe. In other words, the controller **110** can keep the spacecraft **591** starting from the initial state **609** safe with two-thirds likelihood, and the spacecraft **591** starting from the initial state **607** safe with one-third likelihood. Consequently, the set **605** (shown with horizontal pattern fill) is the stochastic controllable set corresponding to a safety likelihood of two-third. The stochastic controllable set **605** therefore includes the initial state **609**, but not the initial state **607**.

[0131] FIG. 6C illustrates an inner-approximation of the stochastic controllable set, according to an embodiment of the present disclosure. A set **626** (shown with vertical pattern fill) is an inner-approximation of a stochastic controllable set **605** (shown with horizontal pattern fill). Any state that lies in the set **626** also lies in the stochastic controllable set **605**. It may be observed from FIG. 6C that neither the set **626** nor the stochastic controllable set **605** covers the entire complement of the keep-away set **601** (shown in grey without pattern fill).

[0132] According to an embodiment, the inner approximation of the stochastic controllable set may correspond to a robust controllable subset. For any state of the spacecraft **591** within the robust controllable subset, there exists a control command maintaining the state of the spacecraft **591** within the robust controllable subset for known or admissible future states of a nominal trajectory.

[0133] FIG. 7 shows a schematic **700** for determining the robust controllable set by leveraging the robust controllable set, according to an embodiment of the present disclosure. Various computational algorithms can be used for computing the robust controllable sets. However, these algorithms are not directly applicable to the problem with the presence of unbounded, stochastic uncertainties **402** including the probabilistic distribution of the process and/or the measurement noise. This is because the computational algorithms are applicable to a problem with presence of bounded, deterministic uncertainties.

[0134] To mitigate this problem, the unbounded stochastic uncertainties **701** are transformed into bounded deterministic uncertainties **703** with a predefined likelihood specifying a range of values on the unbounded stochastic uncertainties **701**. The transformation of block **701** to **703** by converting the unbounded stochastic uncertainty in actuation and measurement to non-stochastic bounded uncertainties that lie in the sets $\epsilon_{act}, \epsilon_{meas}$ are chosen such that the likelihood of the actuation and measurement uncertainty lying in these sets is no smaller than

$$\frac{1}{\alpha_{act}^M}, \frac{1}{\alpha_{meas}^M}$$

respectively, where M is a number of time steps for which safety must be guaranteed during an abort and $\alpha_{act}, \alpha_{meas}$ are selected such that the first likelihood $\beta = \alpha_{act} \alpha_{meas}$. Next, the computational algorithms can be used to compute the robust controllable set **707**. Further, based on the robust controllable sets **705**, robust controllable sets **707** are determined. Specifically, in an embodiment, the inner-approximation of the stochastic controllable set can be determined based on the robust controllable sets **705**. The determination of the inner-approximation of the stochastic controllable set, based on the robust controllable sets **705**, is described below with reference to FIG. 8.

[0135] FIG. 8A shows a block diagram of a method **800** for computing the inner-approximation of the stochastic controllable set, based on the robust controllable sets, according to an embodiment of the present disclosure. At block **801**, the method **800** includes decomposing a complement of a keep-away set (e.g., the complement **303** of the keep-away set **301** as shown in FIG. 3A) as a union of convex sets.

[0136] FIG. 8B shows the complement of the keep-away set **301** decomposed as the union of convex sets, according to an embodiment of the present disclosure. Convex sets **807-821** form the union of convex sets. Specifically, the keep-away set **301** is polytope and the complement of the keep-away set **301** can be expressed as a union of half-spaces. Each of such halfspaces are convex.

[0137] FIGS. 8C-8F collectively show convex halfspaces that together define the complement of the keep-away set **301** used by some embodiments. Sets **823**, **827**, **831**, and **835** are reproductions of the keep-away set **301**. A set **825** includes sets **807**, **809**, and **811**, and a set **829** includes sets **811**, **813**, and **815**. A set **833** includes sets **815**, **817**, and **819**, and a set **837** includes sets **819**, **821**, and **807**. The sets **807-821**, which are convex, are referred to as keep-away complement components.

[0138] Referring back to FIG. 8A, at block **803**, the method **800** further includes computing a robust controllable set for each keep-away complement component. The robust controllable sets are computed for the dynamics of the spacecraft **591** under bounded uncertainty in the actuation and the measurement lying in $\epsilon_{act}, \epsilon_{meas}$. In an embodiment, the computation of the robust controllable sets for any convex set S is done via a recursion backwards-in-time for $k=\{0, 1, \dots, M-1\}$ initialized with $\mathcal{O}_M(\epsilon_{act}, \epsilon_{meas}, S) = S \ominus \epsilon_{meas}$, as given below

$$\mathcal{P}_k(\epsilon_{act}, \epsilon_{meas}, S) = \mathcal{O}_{k+1}(\epsilon_{act}, \epsilon_{meas}, S) \ominus (-\epsilon_{meas}) \ominus (B\epsilon_{act}) \ominus (-A\epsilon_{meas})$$

$$\mathcal{O}_k(\epsilon_{act}, \epsilon_{meas}, S) = (S \ominus (-\epsilon_{meas})) \cap \{z \mid Az \in \mathcal{P}_k(\epsilon_{act}, \epsilon_{meas}, S) \oplus (-B^TV)\}$$

[0139] In the above equations, computational geometry operations $\oplus, \ominus$ refer to the Minkowski sum and the Pontryagin difference operations respectively, and $S = \bigcup_i \text{KeepOut}_i$ is selected for every index i.

[0140] At block **805**, the method **800** further includes computing a union of the robust controllable sets to obtain the inner-approximation of the stochastic controllable set for the complement of the keep-away set, one for each index i. The inner-approximate guarantees the control invariance with the first likelihood of the unbounded stochastic uncertainty, over the horizon of M time steps. In other words, there exists an abort control law that that can steer the

spacecraft **591** away from the keep-away set with the first likelihood for M time steps into the future, in the event of the abnormal (post-failure) operation.

[0141] In an embodiment, the robust controllable set and the corresponding abort control law are determined jointly and interdependently using a single computation of robust controllable set. The computation of the robust controllable set automatically generates a set of control actions for each state that maintains the spacecraft **591** within the robust controllable set. The set of control actions are characterized by the spacecraft dynamics, the control constraints, and the robust control set. The abort control law thus becomes selecting a control action from the set of control actions at each time step.

[0142] According to an embodiment, the abort control law to be applied in the event of the abnormal (post-failure) operation at time T is given by the following set-valued map defined for $t=T, T+1, \dots, T+M$

$$\textcircled{?} = (u \in \mathcal{V} | \textcircled{?} +$$

$$Bu \in (\mathcal{O}_{t+1}(\textcircled{?}, \varepsilon_{\text{meas}}, \mathcal{X}_{\text{KeepOut}}) \ominus (-\varepsilon_{\text{meas}}) \ominus (B\varepsilon_{\text{act}}) \ominus (-A\varepsilon_{\text{meas}})).$$

$\textcircled{?}$ indicates text missing or illegible when filed

[0143] The above equation defines an admissible set $\mathcal{V}_{t,\text{eff}}$ of abort control commands to be applied at time t within a limited available actuation $\mathcal{V}$, a subset of actuation available under normal operation $\mathcal{U}$, such that the spacecraft **591** can be steered away from the keep-away set with likelihood no smaller than the first likelihood

$\beta = \alpha_{\text{act}} \alpha_{\text{meas}}$. By construction, the set $\mathcal{V}_{t,\text{eff}}$ is a polytope.

[0144] Further, the nominal control law, which is applied during the normal operation, steers the spacecraft **591** to the target **593** in the keep-away set, while ensuring that the spacecraft **591** stays within the robust controllable sets and all the necessary state and input constraints are satisfied by the nominal control law. An example of such a nominal control law solves the optimization problem given by equation (1). In an embodiment, the nominal control law is designed using a model predictive controller (MPC). The MPC is based on an iterative, finite horizon optimization of a model of the dynamics of the spacecraft **591**, a set of objectives of the motion of the spacecraft **591**, and constraints on spacecraft propulsion system and motion. The MPC can anticipate future events to take appropriate control actions. According to an embodiment, the MPC may estimate a sequence of control steps over a prediction horizon acting on the spacecraft **591** under the effect of internal and external forces.

Constraint Tightening for Enforcement of Containment in a Plurality of Robust/Stochastic Controllable Set

[0145] Back to FIG. 5B, some implementations consider two types of uncertainty models in nominal operation **133**, i.e., bounded non-stochastic uncertainty and stochastic uncertainty. In both cases, we can use constraint tightening that uses nominal dynamics **132** to tighten the constraints so that the satisfaction of the resulting constraints **140** and **150** ensures that the nominal trajectory satisfies constraints **141** and **157**, despite the nominal uncertainty **133**.

[0146] When a single robust controllable set $\mathcal{J}$ models **157** and a bounded non-stochastic uncertainty $\mathcal{W}_t$ is used in nominal uncertainty model **133**, then **150** is obtained by standard constraint tightening procedures, i.e.,

$$\mathcal{J}^{\text{tighten}} = \mathcal{J} \ominus (\mathcal{W}_k \oplus \text{Reach}_{k|t}(\mathcal{W}, A)), \quad (25)$$

[0147] Where $\text{Reach}_{k|t}(\mathcal{W}_t, A_c)$ is the forward reach set of the uncontrolled system (15) under uncertainty over the period from $t=t, t+1, \dots, k$.

[0148] When a plurality of (robust/controllable) controllable sets, each represented as a constrained zonotope, must be enforced on the nominal trajectory, then we will have to invoke mixed-integer linear programming to reflect the disjunctive constraints. For example, consider N_D constrained zonotopes $\{\mathcal{C}_i\}_{i=1}^{N_D} = \{\mathcal{J}_i^{\text{tighten}}\}_{i=1}^{N_D}$ where $\mathcal{J}_i$ is the constrained zonotope that corresponds to post-failure safety constraints that must be enforced at some time $k \geq t$ in the design of the nominal trajectory at time t . Using N_D auxiliary continuous variables

$$\zeta_i \in \mathbb{R}^{N_{\zeta_i}}$$

and N_D auxiliary binary variables $\delta_i \in \{0, 1\}$, and for any sufficiently large $M > 0$, the following set of mixed-integer linear constraints is equivalent to the constraint the nominal state at (predicted) time k , given the current time t

$$\forall i \in i = 1, \dots, N_D, \|G_{C,i}\xi_i + c_{C,i} - x_{k|t}\|_{\infty} \leq M(1 - \delta_i) \quad (26)$$

$$\forall i \in i = 1, \dots, N_D, A_{C,i}\xi_i = b_{C,i}, \|\xi_i\|_{\infty} \leq 1, \quad (27)$$

$$\sum_{i=1}^{N_D} \delta_i \geq 1. \quad (28)$$

[0149] Such constraints can be easily solve by off-the-shelf solvers like GUROBI.

[0150] Chance constraint enforcement of constrained zonotopic inner-approximations of the stochastic controllable set can be accomplished by enforcing that the mean of the perturbed state lies inside the Pontryagin difference of the constrained zonotopic inner-approximations of the stochastic controllable set and a convex and compact set that has the necessary probability mass that is symmetric about some point. This corresponds to the well-known ellipsoidal approximation of chance constraints.

[0151] Similarly to (26)-(28), one can also use mixed-integer linear constraints to enforce containment in plurality of stochastic controllable sets.

[0152] Many modifications and other embodiments of the disclosure set forth herein will come to mind to one skilled in the art to which these disclosures pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. It is to be understood that the disclosures are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descrip-

tions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

[0153] FIG. 9 is a schematic diagram illustrating some components used for implementing the methods and the systems of the present disclosure. For example, a computer 900 can be adapted for controlling the motion of the spacecraft 101 in the multi-object celestial system while avoiding the unauthorized entry into the keep-away zone 107 during the normal and the abnormal operation of the spacecraft. A CPU or processor(s) 901 can be connected via a bus system 903 to a memory 905, input/output devices 907 and a communication interface 909. Also, connected to the bus system 903 can be a storage device 911, a control interface 913, display interface 915, and an external interface 917.

[0154] The external interface 917 can be connected to an expansion memory 919, vehicle parameters 921 (i.e. spacecraft specifications, thruster specifications, size, weight, etc.), initial orbit data 923 (i.e. time, date, parameters including altitude, inclination, eccentricity, etc.), target orbit data 925, and other orbit data 927 (i.e. unique orbit data). The bus system 903 can also connect a control interface 929, an output interface 931, a receiver 933 and a transmitter 935. Further, the bus system 903 can connect a GPS receiver module 937 to a GPS 939. The computer 900 includes an orbit maintenance module 941. The orbit maintenance module 941 may output thruster commands 943. The orbit maintenance module 941 includes a transfer orbit generator 945, a feedback gain module 947, a feedback controller 949, and a thruster command generator 951.

[0155] The computer 900 can be a server or a desktop, a laptop, a mobile or other computer device or system with one or more processors 901. The processor 901 may be a central processing unit adapted for accessing code in the form of the transfer orbit generator 945 in the memory 905 or storage device 911 of the computer 900 (or in an expansion memory 919). Contemplated are external storage devices if further required depending upon the specific design and aspect of an intended hardware and goal implementation, according aspects related to the systems and the methods of the present disclosure. For example, the computer 900 can be used to implement the steps of the systems and methods, where the memory 905, and/or storage device 911 can store data.

[0156] The stored data in the memory 905 can include executable modules, vehicle data and historical space data. For example, the vehicle data can include specifications of the spacecraft, dimensions, weight, performance data under varied conditions including gravitation forces, and other perturbations, i.e. complex motion(s) of a massive body subject to forces other than the gravitational attraction of a single other massive body in space.

[0157] Further, the vehicle data can include data related to aspects related to vehicle dynamics associated with one or more of the multi-variables, i.e. (1) unusual orbital charac-

teristics of a celestial body, i.e. a natural object which is located outside of Earth's atmosphere, such as the Moon, the Sun, an asteroid, planet, or star; (2) unusual orbital motion the celestial body; (3) celestial body's unusually close orbit around another celestial body; and (4) other known perturbations. The space data can include data related to celestial body(s) system, past missions to celestial body(s) and any other data related to space, the spacecraft and planning orbital designs to other celestial bodies in the universe. For example, the space data can include data about the moons of celestial body(s), such as characteristics of celestial body(s) that can be taken into consideration in developing orbital designs from an initial celestial body(s) orbit to a similar target celestial body(s) orbit.

[0158] The processor 901 of the computer 900 may be two or more processors depending upon the specific application. For example, some steps may require a separate processor to ensure a specific processing time or processing speed associated with the systems and methods of the present disclosure. The receiver 933 or input interface can receive space data that may be up-to-date space data, obtained from either an Earth Mission Control Center or sensors associated with the spacecraft, or some other location, after the stored historical space data stored in the memory 905. The receiver 933 and the transmitter 935 can provide a wireless venue for receiving and sending data to, for example, an Earth Mission Control Center, or some other destination. The GPS receiver module 937 connected to the GPS 939 can be used for navigation related aspects. The computer 900 may further include external devices, control interfaces, displays, sensors, machines, etc., that are contemplated for uses related to the systems and methods of the present disclosure.

[0159] FIG. 10 is a schematic illustrating by non-limiting example a computing apparatus for implementing the methods and the systems of the present disclosure. The computing device 1000 can include a power source 1001, a processor 1003, a memory 1005, a storage device 1007, all connected to a bus 1009. Further, a high-speed interface 1011, a low-speed interface 1013, high-speed expansion ports 1015 and low speed connection ports 1017, can be connected to the bus 1009. In addition, a low-speed expansion port 1019 is in connection with the bus 1009. Further, an input interface 1021 can be connected via the bus 1009 to an external receiver 1023 and an output interface 1025. A receiver 1027 can be connected to an external transmitter 1029 and a transmitter 1031 via the bus 1009. Also connected to the bus 1009 can be an external memory 1033, external sensors 1035, machine(s) 1037, and an environment 1039. Further, one or more external input/output devices 1041 can be connected to the bus 1009. A network interface controller (NIC) 1043 can be adapted to connect through the bus 1009 to a network 1045, wherein data or other data, among other things, can be rendered on a third-party display device, third party imaging device, and/or third-party printing device outside of the computer device 1000.

[0160] The memory 1005 can store instructions that are executable by the computer device 1000, historical data, and any data that can be utilized by the methods and systems of the present disclosure. The memory 1005 can include random access memory (RAM), read only memory (ROM), flash memory, or any other suitable memory systems. The memory 1005 can be a volatile memory unit or units, and/or

a non-volatile memory unit or units. The memory **1005** may also be another form of computer-readable medium, such as a magnetic or optical disk.

[0161] The storage device **1007** can be adapted to store supplementary data and/or software modules used by the computer device **1000**. For example, the storage device **1007** can store historical data and other related data as mentioned above regarding the present disclosure. Additionally, or alternatively, the storage device **1007** can store historical data like data as mentioned above regarding the present disclosure. The storage device **1007** can include a hard drive, an optical drive, a thumb-drive, an array of drives, or any combinations thereof. Further, the storage device **1007** can contain a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid-state memory device, or an array of devices, including devices in a storage area network or other configurations. Instructions can be stored in an information carrier. The instructions, when executed by one or more processing devices (for example, the processor **1003**), perform one or more methods, such as those described above.

[0162] The computing device **1000** can be linked through the bus **1009**, optionally, to a display interface or user Interface (HMI) **1047** adapted to connect the computing device **1000** to a display device **1049** and a keyboard **1051**, wherein the display device **1049** can include a computer monitor, camera, television, projector, or mobile device, among others. In some implementations, the computer device **1000** may include a printer interface to connect to a printing device, wherein the printing device can include a liquid inkjet printer, solid ink printer, large-scale commercial printer, thermal printer, UV printer, or dye-sublimation printer, among others.

[0163] The high-speed interface **1011** manages bandwidth-intensive operations for the computing device **1000**, while the low-speed interface **1013** manages lower bandwidth-intensive operations. Such allocation of functions is an example only. In some implementations, the high-speed interface **1011** can be coupled to the memory **1005**, the user interface (HMI) **1047**, and to the keyboard **1051** and the display **1049** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **1015**, which may accept various expansion cards via the bus **1009**. In an implementation, the low-speed interface **1013** is coupled to the storage device **1007** and the low-speed expansion ports **1017**, via the bus **1009**. The low-speed expansion port **1017**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet) may be coupled to one or more input/output devices **1041**. The computing device **1000** may be connected to a server **1053** and a rack server **1055**. The computing device **1000** may be implemented in several different forms. For example, the computing device **1000** may be implemented as part of the rack server **1055**.

Exemplary Embodiments: Robust Model Predictive Control

[0164] FIG. 11A shows a schematic of vehicle **1120a** including the controller **1100**, according to some embodiments of the present disclosure. As used herein, the vehicle **1120a** can be any type of wheeled vehicle, such as a passenger car, bus, or rover. Also, the vehicle **1120a** can be an autonomous or semi-autonomous vehicle. For example,

some embodiments control the motion of the vehicle **1120a**. Examples of the motion include the lateral motion of the vehicle controlled by a steering system **1103** of vehicle **1120a**. In one embodiment, the steering system **1103** is controlled by the controller **1100**. Additionally, or alternatively, the steering system **1103** can be controlled by a driver of the vehicle **1120a**.

[0165] The vehicle can also include an engine **1106**, which can be controlled by the controller **1100** or by other components of the vehicle **1120a**. Vehicle **1002a** can also include one or more sensors **1104** to sense the surrounding environment. Examples of the sensors **1104** include distance range finders, radars, lidars, and cameras, either mounted on the vehicle or on a road-side unit (RSU). The Vehicle **1120a** can also include one or more sensors **1105** to sense its current motion quantities and internal status. Examples of the sensors **1105** include global positioning system (GPS), accelerometers, inertial measurement units, gyroscopes, shaft rotational sensors, torque sensors, deflection sensors, pressure sensors, and flow sensors. The sensors provide information to the controller **1100**. The vehicle can be equipped with a transceiver **1107** enabling communication capabilities of the controller **1100** through wired or wireless communication channels.

[0166] FIG. 11B shows a schematic of interaction between the controller **1100** and controller **1120** of vehicle **1120a**, according to some embodiments of the present disclosure. For example, in some embodiments, the controllers **1120** of the vehicle **1120a** are a steering controller **1125** and a brake/throttle controller **1130** that control the rotation and acceleration of the vehicle **1120a**. In such a case, the controller **1100** outputs control inputs to the controllers **1125** and **1130** to control the state of the vehicle **1120a**. Controllers **1120** can also include high-level controllers, e.g., a lane-keeping assist controller **1135** that further processes the control inputs of the controller **1100**. In both cases, controller **1120** uses the outputs of controller **1100** to control at least one actuator of the vehicle, such as the steering wheel and/or the brakes of vehicle **1120a**, in order to control the motion of vehicle **1120a**.

[0167] In various embodiments, the controller **1100** is a feedback controller configured for model predictive control that determines the robust controllable set for maintaining the state of the vehicle, starting from the current state of the system, within a prediction horizon defining a number of future control steps. Additionally or alternatively, the controller **1100** can be arranged on a drone and/or a remote server controlling the drone, as illustrated in FIG. 11C.

[0168] FIG. 11C shows a schematic of an embodiment **1100c** where an unmanned aerial vehicle (UAV) is controlled using the controller **1100** implemented on a server **1110c**, according to some embodiments of the present disclosure. The server **1110c** executes the steps of the methods FIG. 1A described earlier. A vehicle **1120c** equivalent is a UAV. The vehicle **1120c** may transmit data **1130c** of its own state and/or environmental data back to the server **1110c**. In some embodiments, a motion plan **1155c** based on joint tracking of vehicle **1120c** and a map state as generated by controller **1100** is updated continuously as additional information is gathered on the disturbance and the environment. The information on the disturbance and the environment may be gathered by the vehicle **1120c** itself, or by any third-party sensors **1131c**, **1132c**, or **1133c**. The sensors **1131c**, **1132c**, and **1133c** may be physical sensors, such as

LiDAR measuring wind speeds, or non-physical sensors, such as meteorological forecasts of the wind speeds or ocean currents, depending on the vehicle 1120c considered.

[0169] The following description provides exemplary embodiments only and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the following description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing one or more exemplary embodiments. Contemplated are various changes that may be made in the function and arrangement of elements without departing from the spirit and scope of the subject matter disclosed as set forth in the appended claims.

[0170] Specific details are given in the following description to provide a thorough understanding of the embodiments. However, understood by one of ordinary skill in the art can be that the embodiments may be practiced without these specific details. For example, systems, processes, and other elements in the subject matter disclosed may be shown as components in block diagram form in order not to obscure the embodiments in unnecessary detail. In other instances, well-known processes, structures, and techniques may be shown without unnecessary detail in order to avoid obscuring the embodiments. Further, like reference numbers and designations in the various drawings indicate like elements.

[0171] Also, individual embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process may be terminated when its operations are completed, but may have additional steps not discussed or included in a figure. Furthermore, not all operations in any particularly described process may occur in all embodiments. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, the function's termination can correspond to a return of the function to the calling function or the main function.

[0172] Furthermore, embodiments of the subject matter disclosed may be implemented, at least in part, either manually or automatically. Manual or automatic implementations may be executed, or at least assisted, through the use of machines, hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware or microcode, the program code or code segments to perform the necessary tasks may be stored in a machine readable medium. A processor(s) may perform the necessary tasks.

[0173] Various methods or processes outlined herein may be coded as software that is executable on one or more processors that employ any one of a variety of operating systems or platforms. Additionally, such software may be written using any of a number of suitable programming languages and/or programming or scripting tools, and also may be compiled as executable machine language code or intermediate code that is executed on a framework or virtual machine. Typically, the functionality of the program modules may be combined or distributed as desired in various embodiments.

[0174] Embodiments of the present disclosure may be embodied as a method, of which an example has been

provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts concurrently, even though shown as sequential acts in illustrative embodiments.

[0175] Although the present disclosure has been described with reference to certain preferred embodiments, it is to be understood that various other adaptations and modifications can be made within the spirit and scope of the present disclosure. Therefore, it is the aspect of the appended claims to cover all such variations and modifications as come within the true spirit and scope of the present disclosure.

1. A feedback controller for controlling an operation of a mechanical system subject to uncertainty on a state of the operation of the system, the feedback controller comprising: at least one processor; and a non-transitory memory having instructions stored thereon that, when executed by the at least one processor, cause the feedback controller to:

- collect a feedback signal indicative of a current state of the operation of the system subject to constraints and current uncertainty on the current state of the operation of the system, wherein the current uncertainty lies in a symmetric bounded set and includes one or a combination of uncertainty on dynamics of the system, uncertainty on a state of the system, uncertainty on control commands to the system, and uncertainty on constraints on the state of the operation of the system;

- determine a robust controllable set for maintaining the state of the system employing closed-form expressions on a constrained zonotope defining the constraints on the state of the operation of the system and an affine transformation of the symmetric bounded set inclosing the current uncertainty into space of the constrained zonotope, wherein the closed-form expressions include a closed-form approximation of a Pontryagin difference between the constrained zonotopic representation and the zonotopic transformation of the symmetric bounded set;

- determine a control command for controlling the operation of the system subject to the robust controllable set; and

- submit the control command to an actuator of the system causing a change in the state of the operation of the system.

2. The feedback controller of claim 1, wherein the closed-form approximation of the Pontryagin difference is an inner approximation determined by scaling the affine transformation and equality constraints describing the constrained zonotope as a constrained zonotopic minuend based on characteristics of a convex, compact, and symmetric subtrahend.

3. The feedback controller of claim 2, wherein the scaling is determined by the solution of a collection of linear equations determined by the affine transformation, the equality constraints describing the constrained zonotopic minuend, and the characteristics of the convex, compact, and symmetric subtrahend.

4. The feedback controller of claim 2, wherein the symmetric bounded set is transformed into the space of the constrained zonotope using higher-dimensional affine transformation.

5. The feedback controller of claim 2, wherein the closed-form approximation of the Pontryagin difference is obtained

by computing the Pontryagin difference between a polyhedral outer approximation of the constrained zonotopic minuend and the subtrahend intersected with a translation of the constrained zonotopic minuend by a point of symmetry of the subtrahend.

6. The feedback controller of claim 5, wherein the polyhedral outer approximation of the constrained zonotopic minuend is determined by a collection of halfspaces determined by a closed-form collection of feasible solutions to a Lagrangian dual of an optimization problem describing a feasibility check of containment of a trajectory of the operation of the mechanical system in the constrained zonotope.

7. The feedback controller of claim 1, wherein the feedback controller is configured for fault-tolerant control by tightening the constraints represented by the constrained zonotope and maintaining a nominal trajectory of the operation of the mechanical system within the robust controllable set approximated using the constrained zonotope.

8. The feedback controller of claim 7, wherein a containment of the nominal trajectory in a constrained zonotopic inner-approximation of the robust controllable set is given by tightening the constraints based on nominal dynamics, nominal uncertainty model, and the constrained zonotope.

9. The feedback controller of claim 8, wherein tightening the constraints is further based on post-failure dynamics, post-failure safety constraints, and a post-failure uncertainty model.

10. The feedback controller of claim 7, wherein the robust controllable set is determined from a stochastic controllable set, by enforcing the constraints as containment in a second non-stochastic constrained zonotope that inner-approximates the Pontryagin difference between a first constrained zonotope and a convex, compact, and symmetric set based on a nominal uncertainty model that contains a given probability mass.

11. The feedback controller of claim 1, wherein the mechanical system is a spacecraft, and wherein the feedback controller is a fault-tolerant controller is configured to perform an abort-safe control under the uncertainty.

12. A spacecraft for moving in a multi-object celestial system while avoiding an unauthorized entry into a keep-away zone during a normal and an abnormal operation of the spacecraft, wherein the normal operation includes moving towards a target in the keep-away zone, and wherein the abnormal operation includes one or a combination of a failure to receive an authorization to enter the keep-away zone and a failure of at least one component of the spacecraft, comprising:

the feedback controller of claim 1;

a set of thrusters configured to change the state of a spacecraft according to a sequence of control commands produced by the feedback controller of claim 1;

a set of sensors configured to produce measurements indicative of the state of the spacecraft; and

a circuitry configured to detect the abnormal operation of the spacecraft.

13. The spacecraft of claim 12, wherein the feedback controller is configured to

execute, during the normal operation of the spacecraft, a nominal control law subject to constraints on maintaining a state of the spacecraft within a union of a plurality of control invariant sets of values of the state of the spacecraft that partially or completely enclose the keep-

away zone, wherein the state of the spacecraft includes a location of the spacecraft and at least one or a combination of a velocity and an acceleration of the spacecraft, wherein each of the plurality of control invariant sets is determined using constrained zonotopes computational geometry such that when the state of the spacecraft is within a control invariant set there is a control command produced by the nominal control law that maintains the state of the spacecraft within the control invariant set despite internal and external forces acting on the spacecraft; and

execute, upon detecting the abnormal operation of the spacecraft, an abort control law associated with the control invariant set including a current state of the spacecraft, wherein at least some different abort control laws are associated with at least some different control invariant sets, and wherein the abort control law is jointly and interdependently determined for the corresponding control invariant set to produce abort control commands moving the spacecraft while avoiding the keep-away zone for any state within the corresponding control invariant set.

14. The spacecraft of claim 13, wherein each of the control invariant sets is a stochastic reachable set determined for a possible abnormal operation defined by the first likelihood of the unbounded stochastic uncertainty.

15. The spacecraft of claim 13, wherein each of the control invariant sets is a robust control invariant set determined for the nominal control law with a bounded non-stochastic uncertainty corresponding to the first likelihood on the unbounded stochastic uncertainty.

16. The feedback controller of claim 1, wherein the feedback controller is configured for model predictive control that determines the robust controllable set for maintaining the state of the system, starting from the current state of the system, within a prediction horizon defining a number of future control steps.

17. The feedback controller of claim 16, wherein the system is a vehicle.

18. A method for feedback control of an operation of a mechanical system subject to uncertainty on a state of the operation of the system, wherein the method uses a processor coupled with stored instructions implementing the method, wherein the instructions, when executed by the processor carry out steps of the method, comprising:

collecting a feedback signal indicative of a current state of the operation of the system subject to constraints and current uncertainty on the current state of the operation of the system, wherein the current uncertainty lies in a symmetric bounded set and includes one or a combination of uncertainty on dynamics of the system, uncertainty on a state of the system, uncertainty on control commands to the system, and uncertainty on constraints on the state of the operation of the system;

determining a robust controllable set for maintaining the state of the system employing closed-form expressions on a constrained zonotope defining the constraints on the state of the operation of the system and an affine transformation of the symmetric bounded set inclosing the current uncertainty into space of the constrained zonotope, wherein the closed-form expressions include a closed-form approximation of a Pontryagin difference

between the constrained zonotopic representation and the zonotopic transformation of the symmetric bounded set;
determining a control command for controlling the operation of the system subject to the robust controllable set; and
submitting the control command to an actuator of the system causing a change in the state of the operation of the system.

19. The method of claim **18**, wherein the closed-form approximation of the Pontryagin difference is an inner approximation determined by scaling the affine transformation and equality constraints describing the constrained zonotope as a constrained zonotopic minuend based on characteristics of a convex, compact, and symmetric subtrahend.

20. The method of claim **18**, wherein the scaling is determined by the solution of a collection of linear equations determined by the affine transformation, the equality constraints describing the constrained zonotopic minuend, and the characteristics of the convex, compact, and symmetric subtrahend.

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